A STEP-BY-STEP GUIDE FOR BIOTIC-ABIOTIC RATE-OF-CHANGE ANALYSIS

Integrating Biological and Carbonate Chemistry Time Series for Ocean Acidification Assessment

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ABSTRACT

Ocean acidification is altering marine ecosystems worldwide, yet robust comparison of biological responses across observing systems remains challenging because biological time series differ substantially in structure, sampling design, informativeness resolution, and temporal coverage. This step-by-step guide presents a standardized methodological framework and accompanying automated workflow for integrating long-term biotic and abiotic observations into comparable rates of change for ocean acidification assessment.

Building on the conceptual framework proposed by Widdicombe et al. (2023), the guide provides practical recommendations for preprocessing biotic and abiotic datasets, including harmonization of taxonomy, units, temporal representation, missing values, quality control, interpolation, and carbonate-system preparation. It then describes a reproducible analytical workflow in which abiotic time series are deseasoned and summarized as continuous long-term rates of change, whereas biological trajectories are segmented into internally consistent linear phases from which normalized biotic rates are estimated.

The resulting paired biotic-abiotic rates of change provide a common basis for comparing responses among biological groups, abiotic drivers, monitoring sites, and ecosystems.

The accompanying open-source R workflow automates input-structure and internal-consistency checks, selected completeness screening, aggregation, segmentation, rate estimation, and generation of standardized outputs while preserving user control over biological interpretation and data preparation.

Together, the guide and software are intended to support consistent application of the framework, facilitate identification of candidate biotic indicators of ocean acidification, and improve reproducibility and comparability among regional and global assessments.

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I. INTRODUCTION

Ocean acidification (OA) is changing seawater carbonate chemistry through the uptake of anthropogenic CO_2 by the ocean (Gruber et al., 2019). Detecting biological responses to these changes remains difficult because biological observations are less standardized than carbonate chemistry observations and are strongly shaped by scope and local ecological context.

Over the past two decades, substantial progress has been made in standardizing carbonate chemistry measurements, abiotic time-series analysis, and ocean acidification observing practices (Dickson et al., 2007; Sutton et al., 2022; UNESCO/IOC, 2019). Comparable progress has been slower for biological observations. Biological time series are often collected using different sampling designs, taxonomic conventions, measurement units, depth ranges, temporal windows, and quality-control procedures (O'Brien et al., 2017). Even when long-term biological observations are available, these structural differences can prevent direct comparison among sites, ecosystems, and biological groups.

Recognizing this challenge, Widdicombe et al. (2023) proposed a conceptual framework for detecting and comparing ocean acidification impacts by relating rates of biotic change to rates of change in carbonate chemistry. That framework identifies biologically relevant response families, including calcification, primary and secondary production, community structure, biodiversity, and genetic adaptation. Its practical implementation, however, requires a standardized method for preparing heterogeneous biological time series and converting them into rate estimates that can be compared with abiotic change.

This guide was developed to bridge that gap. It translates the conceptual framework into a practical workflow for integrating biotic and abiotic time series (**Figure 1**). It provides a step-by-step workflow for preparing biotic and abiotic time series, and estimating biotic and abiotic rates of change. The objective is not to maximize the number of usable time series, but to prevent structurally invalid comparisons. A time series should enter the rate-of-change analysis only when its biological meaning, temporal representation, observational unit, and abiotic context are sufficiently clear.

The guide begins with data preparation because the validity of the rate estimates depends on decisions made before the analytical workflow is applied. For biological time series, these decisions include classification by type of the biological organization level, assessment of dataset informativeness, definition of biological groups, traits, observational units and time-series units, harmonization of names and units, treatment of zeros and missing values, construction of complete temporal grids, quality control before interpolation, and conservative

handling of suspected artifacts. For abiotic time series, the guide covers variable selection, assessment of temporal and spatial suitability, harmonization of variables and units, quality-control filtering, carbonate-system completeness, depth structure, and seasonality. The analytical workflow then estimates abiotic rates of change from deseasoned trends, segments biological trajectories and calculates normalized biotic rates of change. Finally, it pairs each biological segment with the corresponding abiotic rate estimated for the same ecological context.

The final part of the guide describes an accompanying automated workflow implemented in R. The tool provides a reproducible implementation of the rate-of-change analysis using standardized biotic and abiotic input files together with an Excel-based configuration file. It automates input-structure and internal-consistency checks, selected completeness screening, abiotic deseasonalisation, biotic segmentation, rate estimation, diagnostic plotting, and generation of paired biotic-abiotic outputs. Implemented in R, the workflow is designed for users with little or no programming experience. Once the input files and configuration settings have been prepared, the complete analysis can be executed through a single launcher script without modifying or writing R code. The tool is intended to reduce analytical inconsistency across applications while making the methodology accessible to a broad user community, including researchers, monitoring practitioners, and environmental experts who are not R and Python programmers. Nevertheless, the software does not replace expert judgment. Ecological interpretation, data preparation, selection of suitable time series, and assessment of biological plausibility remain the responsibility of the analyst.

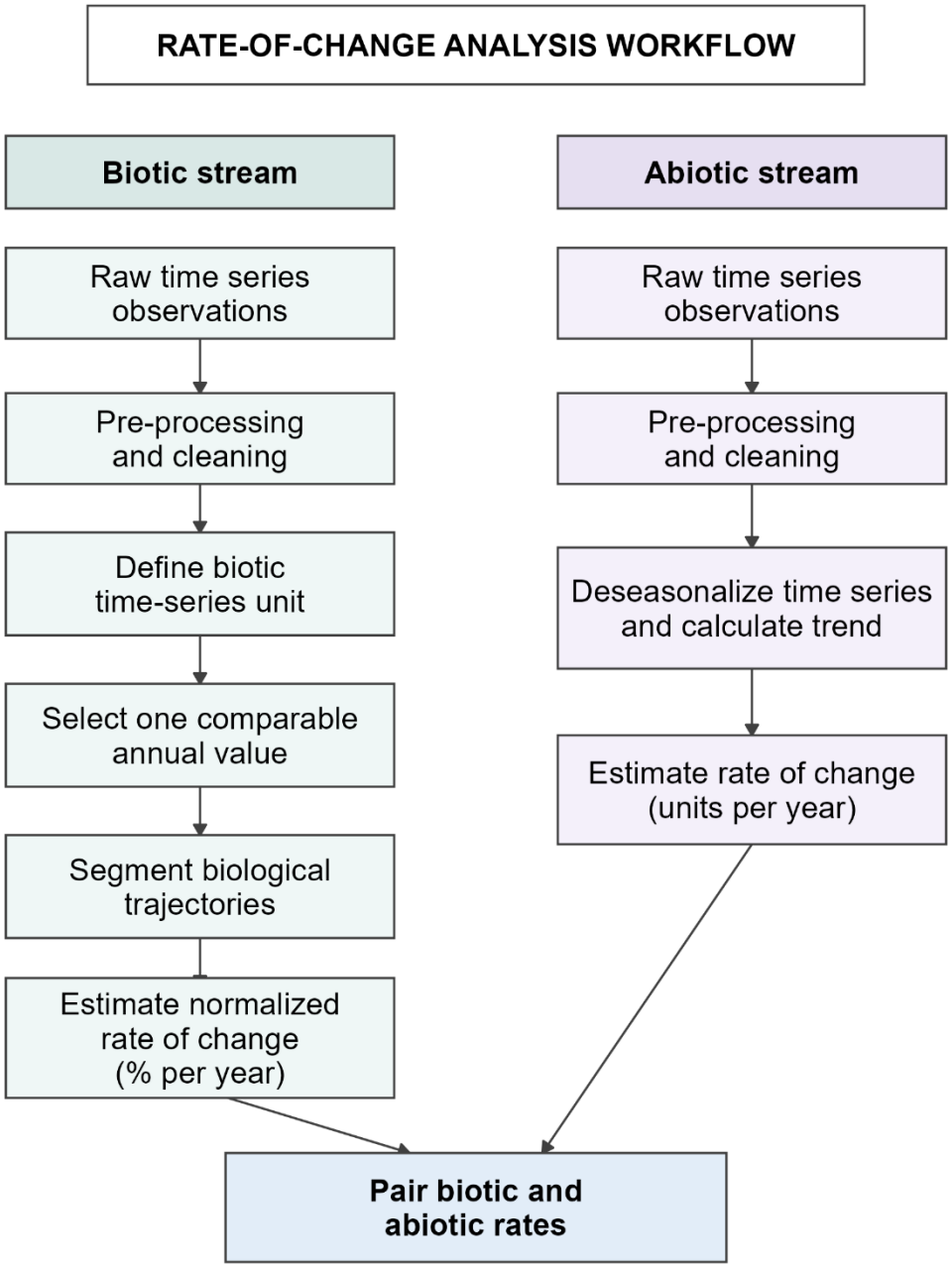


Figure 1. Analytical workflow.

II. TIME SERIES PREPROCESSING AND CLEANING

1. Scope of quality assurance, quality control, and analytical screening

Purpose

Quality assurance, measurement quality control, and analytical screening are distinct processes. Quality assurance comprises procedures used before and during data acquisition to prevent measurement problems, whereas quality control identifies and flags potential problems after observations have been collected. The screening procedures described in this guide subsequently assess whether available observations are sufficiently complete, consistent, and ecologically interpretable for rate-of-change analysis. They do not retrospectively certify measurement quality (U.S. Geological Survey, n.d.).

Implementation

Preserve original observations and provider-assigned quality flags. Record analyst-applied exclusions, recoding, unit conversions, interpolations, extrapolations, and corrections in separate fields or in a linked processing log. If measurement QA/QC is undocumented, consult the data provider where possible and report the resulting limitation; passing the analytical screening in this guide does not establish that the underlying measurements are accurate.

Further QA/QC and metadata resources

General QA/QC concepts and data-quality indicators: U.S. Geological Survey Data Quality Management (https://www.usgs.gov/data-management/manage-quality).
Continuity, method overlap, homogeneity, and long-term metadata: GCOS Climate Monitoring Principles (WMO, 2006).
Carbonate-chemistry sampling, analysis, and quality assurance: Dickson et al. (2007) and the SDG 14.3.1 Indicator Methodology (UNESCO/IOC, 2019).
Platform- and method-specific guidance: GOA-ON OARS QA/QC catalogue (GOA-ON, 2026).

These resources support, but do not replace, the documentation and quality requirements of the original data provider or monitoring programme.

A. BIOTIC TIME SERIES

2. Biotic data general requirements

Substantial efforts have been made to standardize biotic observations (Klein et al., 2019; Muller-Karger et al., 2018). However, biotic time series still rarely follow fully standardized formats

(Bayraktarov et al., 2019; Benson et al., 2022; Bowler et al., 2025), leading to widespread gaps and inconsistencies in taxonomy, space, and time. We identified several recurring data-standardization and quality-control (QC) steps that must be addressed before analysis. If these issues are not resolved, they can strongly affect data processing, reproducibility, and interpretation. Addressing them before analysis reduces uncertainty and helps prevent spurious results.

3. Classify by the level of biological organization

First, classify the type of biological organization being prepared. Long-term monitoring programs and independent time series provide highly diverse biological information. Depending on the organization type, biological data can be classified into three general groups, regardless of the taxonomic identity, functional classification, ecological niche, or measurement method of the biological variable. This classification determines how zeros, missing values, filtering, interpolation, and QC should be handled. **Table 1** provides practical classification guidance.

Table 1. Classification of biotic time series by level of biological organization.

Level of biological organization	**Examples**	**Main question**	**Zero / absence treatment**
Entity-level biological variable	species abundance, genus abundance, relative size of a biological group, biomass, cover	How does the state of a biological entity change through time?	Zero means true absence for any abundance metric (count, cover, density etc.); for measured metrics (e.g. biomass, size) zeros may indicate missing observations or absence of measurement rather than true biological absence
Community-level metric	species richness, Shannon index, trophic richness, compositional change	How does community structure change over time?	These metrics are often calculated from abundance data; therefore, the treatment of zeros should follow the zero/absence rules applied to the underlying entity-level biotic variables

Ecosystem process variable	primary productivity, growth rate, calcification, grazing, respiration	How does a biological process change over time?	Zero may represent a true zero process rate, a value below detection, or a missing observation coded incorrectly as zero. Inspect metadata and measurement context carefully.

4. Time series informativeness and selection of biological traits

To identify biological traits relevant to ocean acidification, we followed the conceptual framework proposed by Widdicombe et al. (2023). That framework identifies five major families of biological traits expected to respond to ocean acidification: (1) calcifying organisms and calcification, (2) autotrophs/mixotrophs and primary production, (3) heterotrophs and secondary production, (4) biodiversity and community structure, and (5) genetic adaptation.

However, the ability to analyse these biological traits depends not only on their ecological relevance, but also on the structure of the available dataset. In practice, the analytical value of biotic time series is determined primarily by sampling frequency and taxonomic resolution. Together, these factors determine the amount of ecological information that can be extracted and the types of biotic metrics that can be estimated. High-frequency data with high taxonomic resolution provide the highest informativeness, allowing analysis of both temporal dynamics and aggregated states at individual and group levels. As frequency or resolution decreases, analytical capacity becomes constrained: low-frequency data limit detection of temporal structure, while low taxonomic resolution restricts interpretation to aggregated system behaviour. At the lowest informativeness level (low sampling frequency and low taxonomic resolution), analyses are generally restricted to group-level, time-integrated metrics.

Consequently, biotic metrics can be broadly divided into two categories:

1. Temporal dynamics (event-resolved metrics), describing within-year biological structure and timing (e.g. bloom initiation, peak timing, or bloom duration), which require high temporal resolution;
2. Aggregated state (time-integrated metrics), describing average biological conditions or system properties over a defined period.

Both metric types can be derived at different levels of biological organization. At the individual level, they describe taxon-specific responses. At the group level, they characterize the collective behaviour of functional groups, communities, or ecosystems (**Figure 2**).

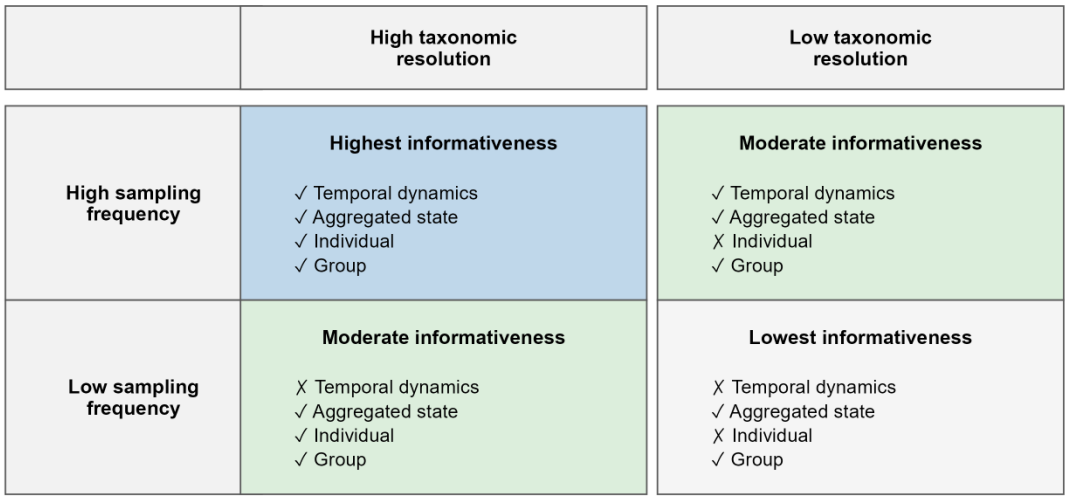


Figure 2. Dataset informativeness based on its structure.

5. Define biological group and its unit, biological trait, and observational unit

Biotic data must be reduced to a consistent analytical representation using the standardized analytical components listed below.

Biological group (bio_group): the biological subject of assessment, whose response is evaluated through time. A biological group may represent a taxonomic group, functional group, size class, community-level metric, ecosystem process, or other biotic variable, depending on the structure of the dataset. Depending on the dataset, a bio_group may correspond to an entity-level biotic variable, a community-level metric, or an ecosystem process variable as defined in **Table 1**.

Biological group category (bio_category): the individual category or subdivision within a biological group (e.g. species within phytoplankton, size classes within copepods, trophic modes within a functional classification). A biological group may also include only one category.

Biological trait (bio_trait): the biological attribute, measurement, classification, or derived metric used to characterize a biological group (e.g. abundance, biomass, calcification rate, pigment concentration, diversity index value).

Observational unit (bio_unit): the smallest analytical unit for which a biological trait value is estimated and a rate of change can be calculated through time. The minimum structure of a bio_unit is site × year × bio_group. Where biological groups are subdivided into categories, the bio_unit becomes site × year × bio_group × bio_category. Additional dimensions such as depth layer, mesh size, habitat, or sampling strata may also be included depending on the data

structure and study objectives. The goal is to obtain a single bio_trait value per year for each defined bio_unit.

Time series unit (ts_unit): the complete biotic trajectory used for rate-of-change estimation. A ts_unit consists of all observations belonging to the same biological group and observational unit through the selected analysis period. In the standard workflow, these observations are typically represented as annual values.

Without explicit separation of the biological subject being assessed (bio_group), its constituent categories (bio_category), the property being quantified (bio_trait), the analytical unit of observation (bio_unit), and the complete trajectory entering the analysis (ts_unit), comparisons across time, taxa, traits, and datasets become ambiguous and difficult to standardize.

6. Harmonize names, taxonomic resolution, units, and missing-value codes

Purpose

The objective of this step is to ensure that observations describing the same biological group, biological category, biological trait, or measurement are represented consistently throughout the entire time series. Without harmonization, artificial variability may be introduced through changes in taxonomy, naming conventions, measurement protocols, units, or missing-value coding. Such inconsistencies can generate spurious trends, distort biodiversity metrics, and reduce comparability among years.

Implementation

Harmonize biological names

Ensure that the same biological group or biological category is represented by a single accepted name throughout the time series. Check for spelling variants, synonyms, historical versus accepted taxonomic names, abbreviations, formatting inconsistencies, extra spaces, and encoding issues.

Example:

Pseudo-nitzschia pungens, *Pseudo nitzschia pungens*, *P. pungens*.

Possible solution:

Refer to metadata describing taxonomic identification methods, taxonomic revisions, and historical changes in taxon reporting. If necessary, prepare a taxonomy metadata table containing the original name, accepted name, taxonomic level, taxon identifier, and correction flag.

Harmonize taxonomic resolution

Identify all taxonomic aggregation levels present in the dataset and determine whether observations are reported consistently through time. Species-, genus-, family-, and higher-level observations should not be treated as equivalent without explicit harmonization.

Example**:**
TOTAL COPEPODS contains observations from Calanoida and Cyclopoida.

Possible solution:

- retain only umbrella categories and sub-groups as separate bio_categories;
- analyse periods separately if taxonomic resolution changed through time;
- exclude inconsistent categories when harmonization is not possible.

Harmonize biotic measurements

a) Harmonize measurement methodology

Check whether sampling design, sampling gear, analytical methodology, spatial coverage, and depth coverage remained sufficiently consistent throughout the time series.

Example:

- mesh 200 µm ≠ mesh 500 µm;
- surface layer ≠ euphotic layer;
- net tow abundance ≠ bottle count abundance.

Possible solution:
If measurements are not directly comparable, treat them as separate biological groups or separate observational units. Methodological differences may become part of the observational-unit definition (e.g., mesh size, depth layer, life stage, size class, or sampling protocol).

Long-term records should be treated as potentially inhomogeneous when station location, sampling depth, sampling time or frequency, gear, instrument, laboratory, calibration procedure, taxonomic resolution, or data-processing method changes. Consistent with the GCOS Climate

Monitoring Principles, document the history of such changes and use an overlap period between old and new methods where possible to evaluate offsets. Do not join records automatically when comparability cannot be demonstrated; retain them as separate observational units, divide the record into methodologically consistent periods, or exclude the incompatible component (WMO, 2006).

b) Harmonize measurement units

Check that identical biological traits are expressed in consistent units before merging observations.

Example:

- cells L^{-1} vs cells m^{-3};
- individuals m^{-3} vs individuals 100 m^{-3};
- chlorophyll µg L^{-1} vs mg m^{-3}.

Possible solution:
Do not merge observations within time series until units have been harmonized and converted where necessary. Record both original and final units, together with conversion factors and assumptions.

Harmonize missing-value codes

Raw datasets often contain multiple representations of missing, invalid, or non-quantitative observations. These values must be converted to a consistent format before any quality control, aggregation, or interpolation.

Example:

NA, N/A, NaN, nd, ND, BDL, blanks, −999, −9999.

Possible solution:
Convert all missing-value indicators to a single standardized NA representation, remove formatting artifacts, and ensure that analytical variables are stored in the correct numeric format before further processing.

Suggested documentation

Record:

- original biological names
- accepted names
- taxonomic aggregation decisions
- unit conversions
- methodological changes
- missing-value recoding rules.

7. Harmonize temporal meaning

Purpose

The objective of this step is to define what one annual biological value represents. The rate-of-change workflow requires one representative bio_trait value per year for each defined bio_unit. However, biological dynamics are often seasonal, episodic, or event-driven, and collapsing them into a single annual value can obscure important ecological information. Therefore, the temporal meaning of the biotic value must be defined explicitly before aggregation, quality control, interpolation, or rate estimation.

Implementation

Each time series must be assigned a consistent temporal representation. The choice should be based on the ecological question, sampling frequency, temporal resolution, the expected biology of the system, and available literature or expert knowledge. Different temporal representations may describe fundamentally different biological phenomena. Consequently, rates of change derived from annual means, seasonal means, bloom characteristics, or event timing metrics should not be interpreted as equivalent biological responses.

Below are listed two main strategies that can be used.

Selection-based representation

A single value is extracted from a fixed and consistent time window or biotic phase.

Examples include:

- abundance in a fixed month;
- biomass during a defined sampling period;

- value at bloom peak;
- bloom start date;
- bloom duration;
- annual maximum.

This approach is appropriate when a specific period or event is ecologically meaningful, or when the available data do not support reliable reconstruction of the full seasonal cycle.

Aggregation-based representation

Values are combined over a defined period to represent the average or integrated biological state.

Examples include:

- annual mean;
- seasonal mean;
- mean abundance during the productive season;
- mean biomass during a selected ecological phase.

This approach is appropriate when the goal is to represent the general state of the biological system over a defined temporal window.

The selected temporal representation must remain constant throughout the entire time series. For example, a time series should not combine annual means in some years with seasonal means, peak values, or single-month observations in other years. Any shift in sampling window, aggregation period, or metric definition can introduce artificial variability that cannot be separated from ecological change.

If the study system has strong seasonality and sampling frequency is sufficient, seasonal time series may be constructed separately. In that case, each season becomes part of the bio_unit, and the final structure becomes, for example:

site × year × season × bio_group

or, when depth is also retained:

site × year × season × layer × bio_group.

If sampling frequency is insufficient to support reliable seasonal separation, a single consistent annual value may provide a more defensible representation of the biological system.

Suggested documentation

Record:

- the selected temporal representation;
- whether the value is selected or aggregated;
- the temporal window used;
- the aggregation function used, if applicable;
- whether seasons or phases were treated as separate ts_units;
- justification based on sampling frequency, ecosystem knowledge, or literature;
- any months, seasons, or periods excluded from analysis.

8. Decide zero vs NA treatment

Purpose

The objective of this step is to determine whether observations recorded as zero represent true biological information or whether they should be treated as missing values. This distinction must be established before constructing the complete temporal grid because zeros and missing values carry fundamentally different ecological meanings and require different treatment during quality control, interpolation, biodiversity calculations, and rate-of-change analysis. Confusing zeros with missing values can introduce artificial trends, bias biodiversity estimates, distort interpolation, and produce misleading rates-of-change.

Implementation

In biotic time series, a value of zero often represents meaningful ecological information. Depending on the dataset and sampling methodology, a zero may indicate observed absence of a biological entity, local disappearance, absence of a life stage, size class, or trait category during sampling, or a valid measured value equal to zero. In contrast, a missing value indicates that the observation is unknown. Before constructing the complete temporal grid, determine the meaning of each empty observation and apply a consistent rule throughout the dataset.

The distinction should be defined explicitly according to the measurement protocol and ecological meaning of the variable:

0 = observed absence or a valid observed value equal to zero;

NA = unknown value or no observation available.

Values below analytical detection limits require case-specific evaluation and should not automatically be treated as either zero or missing. The appropriate treatment depends on the type of biological organization and the biological meaning of the measurement.

For abundance-based observations, zeros frequently represent true ecological information and should usually be retained. These values are essential for detecting biologically meaningful dynamics such as appearance, establishment, decline, and disappearance.

For biomass, productivity, physiological measurements, and biodiversity metrics, a value of zero may be biologically impossible or may represent a measurement problem. In these cases, zeros should be evaluated carefully before being treated as true observations.

Suggested documentation

Record:

- the definition of zero used in the dataset;
- the definition of missing values used in the dataset;
- any assumptions regarding values below detection limits;
- variables for which zeros were converted to NA;
- how values below detection limits were treated.

9. Build a complete temporal grid

Purpose

Biotic observations are often recorded only when a biological group is detected, measured, or reported. As a result, many datasets contain implicit omissions where years, biological groups, seasons, layers, size classes, or other observational units are absent from the table. Such omissions create structural inconsistencies that can distort trend estimation, biodiversity metrics, interpolation procedures, and rate-of-change calculations. The objective of this step is to create a structurally complete time series in which every expected year is represented for every observational unit. This allows true absences, missing observations, taxon appearances, and taxon disappearances to be represented explicitly and treated consistently throughout the workflow.

Implementation

Construct a complete temporal grid containing all expected combinations defining the observational unit (bio_unit) through time.

For example:

site × *year* × *bio_group,*

or

site × *year* × *season* × *layer* × *bio_group*.

All expected combinations should be represented explicitly, even when no value was reported in the original dataset, so that absence and missingness can be distinguished consistently. After the grid has been created, populate missing combinations according to the zero-versus-NA rules defined in the previous step.

Examples:

Observed absence	→ 0
Below detection limit (if treated as absence)	→ 0
No sample collected	→ NA
Variable not measured	→ NA
Unknown value	→ NA

The resulting dataset should contain one record for every expected year and observational unit, regardless of whether the value is observed, absent, or missing.

Suggested documentation

Record:

- the definition of the observational unit;
- rules used to assign zeros and missing values.

10. Apply biological organization-specific completeness screening

Purpose

The objective of this step is to determine whether each time series contains sufficient real information to support reliable estimation of biotic rates of change. The screening criteria (SC) must be selected according to the ecological meaning of the biological organization type, rather than applied uniformly across all variables. This step is not intended to remove natural variability.

Its purpose is to exclude time series that are too sparse, discontinuous, or structurally uninformative to support meaningful trend estimation. SC must be performed before interpolation. Interpolation should only fill small gaps in otherwise informative time series; it should not create information where little or none exists.

The criteria in this section assess fitness for the present analysis. They are not measurement-quality criteria and do not establish whether the underlying observations were collected, calibrated, or analysed correctly.

Implementation

Apply SC separately to each ts_unit, calculate:

- total number of years;
- number of observed years;
- number of missing years;
- proportion of missing values;
- proportion of zero values;
- number of positive observations.

The SC criteria should then be selected according to biological organization type.

Table 2. Recommended screening strategy according to biological organization type.

Biological organization type	**Entity-level abundance variables**	**Entity-level measured variables**	**Ecosystem process variables**	**Biodiversity metrics**
Time series examples	· species abundance · genus abundance · functional-group abundance · size-class abundance · life-stage abundance	· total biomass · ash-free/dry mass · relative size	· primary productivity · growth rate · calcification rate · grazing rate · respiration	· species richness · Shannon diversity · evenness · compositional change · trophic richness
Comment	Abundance series frequently contain	Zeros are usually not interpreted as	These variables usually contain	Applying zero-proportion filters

	both ecological zeros and missing observations	ecological absences and often require evaluation before being retained or converted to NA	missing measurements rather than ecological absences	would bias results because low diversity may itself be ecologically meaningful.
SC should evaluate	·minimum number of observed years ·maximum proportion of missing values ·maximum proportion of zeros ·minimum number of positive observations	·minimum number of observed years ·maximum proportion of missing values	·minimum observed years ·maximum missing proportion	·minimum observed years ·maximum missing proportion
Proposed criteria for ~20-year time series	< 30% missing values < 75% zeros	< 30% missing values	< 30% missing values	< 30% missing values

The proposed thresholds are practical heuristics developed for this workflow. They are intended to identify time series that contain sufficient information for rate-of-change estimation while retaining a broad range of biotic variables. Users may adopt stricter criteria. For comparison, IGMETS trend analyses require at least 80% data availability within a time window (O'Brien et al., 2017).

Suggested documentation

Record:

- SC criteria applied to each biological organization type;
- threshold values used;
- reason for exclusion;
- whether zero-proportion filters were applied.

11. Interpolate remaining missing values

Purpose

The objective of this step is to improve continuity within informative time series by estimating a limited number of missing observations that remain after completeness screening. Interpolation is intended to support trend estimation in otherwise well-sampled records and should not be used to create information where little or none exists. Only time series that passed the type-specific screening criteria should be considered for interpolation. Interpolation rules must be defined before analysis and applied consistently across comparable biotic variables.

Implementation

Interpolation

Apply interpolation separately to each retained time-series unit (ts_unit). Linear interpolation is recommended because it is transparent, reproducible, easy to interpret, and preserves long-term temporal trajectories without introducing unnecessary complexity or artificial variability.

Only missing values (NA) should be interpolated. True zeros should remain unchanged because they represent ecological information rather than missing observations.

Before interpolation, define a maximum number of consecutive missing years that may be filled. Record the number and proportion of interpolated values within every ts_unit. Where interpolation could influence slope or breakpoint selection, compare the results with and without interpolated values and report any material difference.

Interpolation should only be applied when sufficient observed values exist to support a meaningful estimate and when missing observations occur as relatively small gaps within an otherwise informative time series. It should not be used to reconstruct heavily fragmented records, replace structural missingness, or recover time series that failed quality control. Interpolated values are estimates rather than observations and should therefore be interpreted as supporting trend estimation rather than providing new biological information.

Extrapolation

Extrapolation should normally be disabled in the primary analysis because estimated values at the beginning or end of a record can directly influence fitted slopes, segment boundaries, and normalized rates of change. If extrapolation is retained for a clearly justified application, the number of extrapolated endpoints must be limited in advance, all values must be flagged, and sensitivity analyses with and without extrapolation must be reported.

Suggested documentation

To preserve transparency and reproducibility, all estimated values should remain fully traceable. Original observations should be preserved, and quality-control flags should explicitly identify interpolated and extrapolated values.

For example:

- bio_value_original – original observed value;
- bio_value_final – value used in subsequent analyses;
- interpolated_flag – indicates interpolated values;
- extrapolated_flag – indicates extrapolated values.

Record:

- interpolation method used;
- extrapolation method used, if applicable;
- minimum observation threshold required for interpolation;
- interpolation and extrapolation flags.

12. Apply spike/artifact QC only when justified

Purpose

This step is optional and should be applied conservatively. The objective of this step is to identify isolated observations that are likely to represent analytical, processing, transcription, or data-handling artifacts and that could disproportionately influence trend estimation and rate-of-change calculations. Extreme biotic observations should not be removed solely because they are unusual. Large values may reflect genuine ecological phenomena, including blooms, recruitment events, regime shifts, extreme environmental conditions, changes in population structure, or other biologically meaningful processes. Unless evidence suggests otherwise, observations should be assumed to be valid. The justification for this quality-control step arises from the subsequent rate-of-change workflow. Both linear trend estimation and temporal segmentation procedures can be strongly influenced by a single extreme observation. An isolated artifact may generate artificially high rates of change, bias slope estimates and model-fit statistics, introduce false segmentation events, and distort ecological interpretation.

Many biotic time series will not require this step. In most cases, biologically plausible extreme observations should be retained and carried forward into the analysis unchanged.

Implementation

Spike/artifact quality control should only be considered when all of the following conditions are satisfied:

- the observation forms an isolated peak or trough;
- the value is disconnected from both preceding and following observations;
- the magnitude greatly exceeds surrounding values;
- no biotic, abiotic, methodological, or sampling explanation can be identified.

Before considering correction, potential explanations should be investigated, including:

- bloom events;
- recruitment pulses;
- regime shifts;
- extreme environmental conditions;
- changes in sampling design;
- changes in analytical methods;
- taxonomic reclassification;
- altered detection limits;
- other documented ecological processes.

Conventional statistical outlier-detection approaches, such as z-scores, interquartile-range thresholds, Grubbs' tests, or distribution-based outlier detection, are not recommended as default procedures. Biological extremes are frequently ecologically meaningful rather than erroneous, and statistical rarity alone should not be interpreted as evidence of error. The objective is therefore not to identify statistically unusual observations, but to identify implausible artifacts capable of distorting trend estimation and segmentation. We recommend a conservative expert-guided assessment of suspicious observations that considers temporal context, ecological plausibility, sampling design, analytical methodology, and data provenance.

Screening of candidate artifacts

To support objective screening, candidate spikes may be flagged using heuristic criteria such as:

max1 / max2 > threshold

max1 / median > threshold

where:

- max1 = largest observation in the time series;
- max2 = second-largest observation in the time series;
- median = median value of the time series.

Example screening thresholds are:

max1 / max2 > 4

max1 / median > 20

These values are workflow-specific heuristics and should not be interpreted as universally applicable thresholds, and should be interpreted only as screening criteria. They are not statistical tests and should not be treated as evidence that an observation is erroneous. The proposed threshold values are intentionally conservative and are designed to identify exceptionally isolated observations that warrant manual review. They should not automatically trigger correction.

Correction of confirmed artifacts

If an observation is judged to represent an artifact after manual review, it may be replaced with NA and subsequently treated using the same interpolation procedure applied to other retained gaps in the time series. If the observation cannot be confidently classified as an artifact, it should be retained.

Proposed documentation

All corrections should remain fully traceable. Original observations should be preserved, and all modifications should be explicitly flagged.

Keep the original provider quality flag, the analyst screening flag, and the final analysis decision as separate fields. A suspicious value should first be flagged for review; the raw value should never be overwritten.

Recommended fields include:

- bio_value_original – original observed value before interpolation or artifact correction;

- bio_value_before_outlier_qc – value prior to spike/artifact correction;
- bio_value_final – final value used in analysis;
- artifact_corrected_flag – indicates observations modified during spike/artifact quality control.

Record:

- screening criteria used;
- threshold values applied;
- observations flagged for review;
- justification for each correction;
- interpolation method applied after correction, if applicable.

B. ABIOTIC TIME SERIES

13. General requirements

Robust estimation of abiotic rates of change requires careful preparation of abiotic time series before analysis. Detailed recommendations for ocean acidification time-series analysis, including quality control, treatment of missing observations, deseasonalisation, trend estimation, uncertainty assessment, and reporting of long-term abiotic change, are already available (Sutton et al., 2022). Rather than duplicating these established methodologies, this guide adopts them as the abiotic component of the framework. Users are therefore strongly encouraged to follow the recommendations of Sutton et al. (2022) when preparing and evaluating abiotic time series.

For users requiring more detailed guidance, Dickson et al. (2007) provides standard operating procedures and quality-assurance guidance for discrete carbonate measurements; the SDG 14.3.1 methodology provides OA-specific measurement-quality and metadata requirements (UNESCO/IOC, 2019); and the GOA-ON Requirements and Governance Plan explains fitness-for-purpose distinctions between climate-quality and higher-frequency weather-quality observations (Newton et al., 2015). Platform-specific resources are maintained in the GOA-ON OARS QA/QC catalogue (GOA-ON, 2026).

14. Define the abiotic variables

Purpose

The objective of this step is to identify the abiotic variables that represent the abiotic driver of interest and provide sufficient context for interpretation of biological responses. The abiotic variables selected at this stage determine which abiotic rates of change will be estimated and subsequently paired with biotic rates of change.

Implementation

At minimum, identify the primary abiotic variable or variables representing the abiotic pressure under investigation. For ocean acidification studies, this should include carbonate chemistry variables. Whenever possible, at least two carbonate-system parameters should be available, preferably as measured observations, together with temperature and salinity. When only a subset of carbonate-system variables is available, additional parameters may be derived using established carbonate-system calculations (see **Section II.19**). Minimum OA-relevant pairs may include any two of: pH, partial pressure of carbon dioxide (pCO_2), dissolved inorganic carbon (DIC), total alkalinity (TA).

However, carbonate chemistry should not be interpreted in isolation when relevant abiotic co-drivers are available. Biological responses are often influenced by multiple interacting abiotic factors. Therefore, where possible, additional abiotic variables should be included to support interpretation of biotic change and to distinguish OA effects from broader abiotic variability.

Depending on the ecosystem and research question, relevant co-drivers may include: temperature, salinity, dissolved oxygen, nutrients, stratification intensity, freshwater influence, upwelling intensity, circulation indices, or extreme-event metrics. If the research objective focuses on abiotic extremes rather than long-term mean conditions, event-based metrics should be derived from the raw observations and shaped into the time+series format before trend analysis. Examples may include: marine heatwave frequency, marine heatwave duration, hypoxia-event frequency, low-pH event frequency, etc.

When *in situ* observations are unavailable, physical or biogeochemical model products and remote-sensing products may be considered, provided that their spatial and temporal resolution is sufficient to represent the study system.

Suggested documentation

Record:

- data source (in situ observations, model output, remote sensing);
- temporal coverage;

- spatial coverage
- variable definition and units;
- whether the variable was measured, modelled, or derived.

15. Evaluate abiotic contextual suitability

Purpose

The objective of this step is to determine whether the selected abiotic time series can support meaningful trend estimation and rate-of-change analysis. Abiotic data do not need to match the biotic data at every sampling date, but they must represent the same study period and ecological context closely enough to support interpretation of biotic change.

Abiotic time series may extend beyond the biotic observation period and can provide useful context regarding historical variability. However, abiotic rates of change used within the workflow should be estimated using the abiotic observations corresponding to the selected biotic analysis window.

Implementation

Contextual suitability should be evaluated using four main criteria: temporal resolution, temporal coverage, measurement consistency, and representativeness.

Temporal resolution

Sampling frequency should be evaluated relative to local variability, the statistic being estimated, and the required accuracy. Monthly sampling may be sufficient to represent broad seasonal structure in some systems, but it is not a universal minimum. Highly variable coastal systems and analyses targeting extremes may require substantially higher-frequency observations (UNESCO/IOC, 2019).

Lower-frequency observations may still support broad interpretation, but their capacity to represent the seasonal cycle, extremes, and a long-term trend must be demonstrated rather than assumed.

Temporal coverage

The abiotic time series should contain enough observations across the study period to support trend estimation after accounting for seasonal variability. There is no universal minimum time span because suitability depends on sampling frequency, variability, and the strength of the

abiotic trend. However, the abiotic record should overlap with the biotic analysis window. If the overlap is short, discontinuous, or concentrated in only part of the biological record, abiotic rates of change should be interpreted cautiously.

The adequacy of the abiotic time series should also be evaluated in relation to the estimated abiotic rate of change. Slowly changing variables may require substantially longer records to support robust trend estimation than rapidly changing variables (check Trend Detection Time in **Section V.11**).

Measurement consistency

Analytical methods, sensors, calibration procedures, detection limits, and data-processing approaches should remain consistent throughout the record or be sufficiently documented to allow correction. Changes in methodology may introduce artificial shifts that can be mistaken for abiotic change.

Following the GCOS Climate Monitoring Principles, assess the likely impact of a change before combining records, document the complete change history, and use overlap between old and new systems where possible to quantify offsets. A high correlation during an overlap period does not demonstrate agreement. Evaluate absolute differences, time-dependent bias, variance, seasonality, and possible step changes. If comparability cannot be established, split the series or retain the methods as separate variables rather than applying an undocumented correction (WMO, 2006).

Abiotic representativeness

The abiotic observations should represent the conditions experienced by the biological community under investigation. Exact spatial co-location is not always necessary, but the selected abiotic series should correspond to the same water mass, habitat, depth range, or ecological system. Abiotic observations obtained from alternative sources, including biogeochemical models or remote-sensing products, should be evaluated for their ability to represent local conditions adequately.

Abiotic trajectories may exhibit non-linear behaviour over long periods. However, within the present framework, abiotic change is intentionally represented by a single deseasoned long-term rate estimated over the full biotic analysis window. If strong abiotic non-linearity is present, it should be considered during interpretation of the results and may motivate additional exploratory analyses outside the standard workflow.

Suggested documentation

Record:

- temporal resolution of the abiotic series;
- temporal overlap with the biotic analysis window;
- major gaps or discontinuities;
- measurement methods and known methodological changes;
- spatial and depth representativeness.

16. Harmonize variables and units

Purpose

The objective of this step is to ensure that observations describing the same abiotic property are represented consistently throughout the time series. Abiotic data often originate from different laboratories, sensors, analytical methods, repositories, or monitoring programs. Without harmonization, differences in variable names, units, calculation methods, or measurement sources may introduce artificial variability and compromise trend estimation.

Implementation

Harmonize variable names and definitions

Identify all variables describing the same abiotic property and verify whether they are directly comparable.

Examples include:

- Salinity_bottles / Salinity_CTD;
- NO3_UDO / NO3_NO2_USF / NO2_USF;
- measured pH / corrected pH.

Before merging variables, confirm that they represent the same abiotic quantity and that any differences in analytical methodology, corrections, or calculation procedures are understood.

Harmonize units

Abiotic variables are frequently reported in different units, scales, or conventions. Before combining or comparing observations, all measurements should be converted to a common unit system. The selected units should be applied consistently throughout the entire time series.

Evaluate alternative measurements

Whenever multiple measurements exist for the same abiotic property, overlap periods should be examined before variables are merged or used to supplement missing observations.

Recommended evaluations may include:

- visual inspection of paired observations;
- scatterplots and comparison against the 1:1 line;
- correlation analysis (recognizing that high correlation alone does not imply agreement);
- examination of differences between measurements;
- assessment of systematic offsets;
- evaluation of temporal consistency;
- assessment of whether agreement changes among years, methods, sensors, or laboratories.

Particular attention should be paid to whether one source consistently produces higher or lower values than another and whether the relationship remains stable through time.

Based on these evaluations, one of the following strategies may be adopted:

- select a preferred variable;
- establish a hierarchy of preferred and secondary variables;
- merge observations when measurements are sufficiently comparable;
- retain variables separately if important differences remain.

Suggested documentation

Record:

- original variable names;
- harmonized variable names;

- unit conversions;
- measurement sources;
- overlap-comparison results.

17. Apply quality-control filtering

Purpose

The objective of this step is to remove observations that have been flagged as unreliable before they influence carbonate-system reconstruction, trend estimation, or rate-of-change calculations.

Implementation

Abiotic datasets frequently contain quality flags indicating whether an observation passed analytical, calibration, or post-processing quality-control procedures. Before any calculations are performed, review the available metadata and identify the meaning of all quality flags. Observations flagged as invalid, unreliable, or unsuitable for scientific analysis should be excluded from further processing according to the recommendations of the data provider or monitoring programme..

Interpret every flag using the definitions supplied by the original provider; identical numeric codes can have different meanings in different repositories. When combining sources, retain the original flag and create a separate harmonized flag and analysis-decision field. An absent flag or a flag indicating that no QC was performed must not be interpreted as good data. At minimum, distinguish unassessed, good, probably good, probably bad, bad, changed, interpolated, and missing observations. The Ocean Data View quality-flag crosswalk provides mappings among commonly used oceanographic schemes (Ocean Data View, n.d.).

Suggested documentation

Record:

- quality-control flags applied;
- flag definitions;
- variables affected by filtering.

18. Clean missing-value codes

Purpose

The objective of this step is to ensure that missing, invalid, or non-quantitative observations are represented consistently before evaluating carbonate-system completeness or performing subsequent calculations.

Implementation

Abiotic datasets frequently contain multiple representations of missing values. Before assessing carbonate-system completeness, convert all missing-value indicators to a standardized NA representation, remove formatting artefacts, and ensure that analytical variables are stored in the correct numeric format.

Example:

NA, N/A, NaN, nd, ND, BDL, blanks, −999, −9999.

This step should be completed before evaluating completeness, applying carbonate-system calculations, or comparing alternative variables.

Documentation

Record:

- missing-value recoding rules.

19. Verify carbonate-system completeness

Purpose

The objective of this step is to determine whether each abiotic observation contains enough valid information to calculate or interpret carbonate chemistry consistently. This step must be completed before deriving additional carbonate-system variables and before abiotic trend analysis.

Implementation

Identify available carbonate-system variables

First, list all measured or reported carbonate-system variables available in the dataset.

Common carbonate-system variables include:

- pH;
- total alkalinity (TA);

- dissolved inorganic carbon (DIC);
- partial pressure of carbon dioxide (pCO_2);
- fugacity of carbon dioxide (fCO_2);
- carbonate ion concentration (CO_3^{2-});
- bicarbonate ion concentration (HCO_3^-);
- aragonite saturation state (Ωarag);
- calcite saturation state (Ωcalc).

Define valid carbonate-calculation observations

A carbonate calculation should only be performed for observations containing:

- a valid pair of carbonate-system variables;
- valid temperature;
- valid salinity;
- valid depth or pressure, where applicable.

Common valid carbonate-system input pairs include:

- pH + total alkalinity;
- pH + dissolved inorganic carbon;
- total alkalinity + dissolved inorganic carbon;
- pCO_2 + total alkalinity;
- and other carbonate-system parameter pairs supported by the selected carbonate-system solver

Observations lacking the required input variables should remain in the dataset, but derived carbonate variables should be recorded as missing values.

A mathematically valid carbonate-system input pair does not constrain every derived variable equally well. The suitability of a pair must therefore be evaluated for the specific output variable and intended use. Derived variables must be identified explicitly and must not be treated as equivalent to direct measurements.

Calculate carbonate-system variables

For observations with complete and valid input data, carbonate-system variables may be calculated using a recognized carbonate chemistry solver based on equilibrium thermodynamics. Examples include the *seacarb* package for R (Gattuso et al., 2024), PyCO2SYS, CO2SYS, or equivalent implementations.

Record uncertainty for each measured input and, where possible, propagate it to calculated outputs using an established approach such as Orr et al. (2018). Propagation should account for the selected input pair, input uncertainties and covariance where available, equilibrium constants, temperature, salinity, pressure, and nutrient inputs. If uncertainty propagation cannot be performed, state this limitation and identify the derived variables most sensitive to the selected input pair.

Suggested documentation

Record:

- carbonate-system solver or package;
- units of all input and output variables;
- selected carbonate-system constants and calculation options, where applicable;
- → measurement uncertainty for each input variable;
- → whether each variable was measured, modelled, or derived;
- → pH scale and measurement temperature, where applicable;
- → carbonate-system input pair and solver version;
- → uncertainty-propagation method and uncertainty of derived outputs, where available.

20. Define and evaluate depth structure

Purpose

The objective of this step is to determine whether observations should be represented by a single depth-integrated value or by separate depth layers. Abiotic conditions and biological responses often vary substantially with depth, and combining observations from distinct depth zones may obscure biologically meaningful patterns. When abiotically or biologically distinct depth strata exist, depth may become part of the observational unit (bio_unit).

Implementation

Identify biologically relevant depth zones

Begin by identifying the major vertical zones expected to influence the biological process under investigation. Layer definitions should reflect meaningful environmental or ecological boundaries rather than arbitrary depth intervals.

The biological question should guide the initial selection.

Examples:

Phytoplankton	→ surface layer or euphotic zone
Zooplankton	→ surface, euphotic, or depth-specific layers
Benthic organisms	→ bottom layer
Microbial communities	→ microbial depth zone

Evaluate abiotic differentiation with depth

Inspect depth profiles of key abiotic variables to determine whether different depth ranges represent distinct abiotic conditions. The objective is not simply to describe vertical gradients, but to determine whether separate depth layers represent distinct ecological environments that may influence biological responses differently.

Evaluate sampling coverage

Before constructing depth-specific time series, verify that sampling frequency and depth coverage are sufficient within each proposed layer. If observations are sparse within individual layers, a depth-integrated representation may be more robust than maintaining multiple poorly sampled layers.

Suggested documentation

Record:

- selected layer boundaries;
- ecological justification;
- aggregation method used within layers;

- sampling coverage within each layer.

21. Define and evaluate seasonality

Purpose

The objective of this step is to determine whether biotic observations should be represented as a single annual value or as separate seasonal time series. In ecosystems with strong and recurring abiotic seasonality, annual averages may obscure biologically meaningful responses occurring during specific abiotic phases. When abiotic seasonality is pronounced and biological sampling frequency is sufficient, season may be incorporated into the observational unit (bio_unit).

Implementation

Define provisional seasonal boundaries

Begin by defining a provisional seasonal structure based on available literature, ecosystem knowledge, hydrography, or previous studies. Season definitions should be treated as hypotheses that require evaluation rather than fixed assumptions.

Evaluate biotic sampling frequency

Before constructing seasonal biotic time series, determine whether biological observations are sufficiently frequent and consistent to support seasonal aggregation. If biological sampling is sparse or highly irregular, seasonal time series may be unreliable regardless of abiotic seasonality.

Evaluate abiotic distinctiveness of seasons

Use abiotic time series to assess whether the proposed seasons correspond to distinct abiotic conditions. The objective is not simply to describe seasonality, but to determine whether different seasons represent biologically meaningful abiotic contexts.

Relevant variables may include:

- temperature;
- salinity;
- dissolved oxygen;

- nutrients;
- carbonate chemistry variables;
- chlorophyll;
- primary productivity;
- other ecosystem-specific indicators.

Select an appropriate depth layer

Seasonality should be evaluated within the depth range most relevant to the biotic observations under investigation. For example, surface layers are often most appropriate when evaluating phytoplankton communities, primary productivity, or upwelling-driven abiotic variability.

Calculate monthly climatology

Calculate mean abiotic conditions for each month across the full time series to identify recurring seasonal patterns and evaluate whether the proposed seasonal boundaries are environmentally justified. If seasonal boundaries remain uncertain, additional diagnostic approaches may be used, including:

- monthly climatology plots;
- hierarchical clustering;
- month-by-month comparisons.

These methods should be used to support expert interpretation rather than to determine seasonal boundaries automatically.

Define the final biotic time-series structure

The final decision should consider both abiotic seasonality and biotic sampling frequency.

General guidance is:

weak seasonality + sparse biotic sampling	→ one annual biotic value
strong seasonality + sparse biotic sampling	→ seasonal separation usually not reliable; use one annual value or interpret cautiously
weak seasonality + frequent biotic sampling	→ one annual biotic value is usually sufficient
strong seasonality + frequent biotic sampling	→ construct one biotic value per season

When seasonal separation is adopted, season becomes part of the observational unit.

Examples:

site × year × season × bio_group,

or

site × year × season × depth layer × bio_group.

Suggested documentation

Record:

- season definitions;
- literature or ecosystem knowledge supporting the definitions;
- abiotic evidence supporting seasonal separation;
- depth layer used for seasonality evaluation;
- excluded months or periods, if applicable.

III. COMPANION METADATA AND PROCESSING HISTORY

Purpose

The minimum analytical input tables are not sufficient to preserve the information required to evaluate measurement quality, reproduce data preparation, or reuse the resulting time series. Each analysis should therefore include companion metadata and a processing history linked to the biotic and abiotic input files. Metadata and provenance should be treated as part of the analytical record rather than as optional supporting information (Wilkinson et al., 2016).

Implementation

The information may be stored in a metadata workbook, machine-readable table, or processing log within the data/metadata/ directory. Use the original provider's metadata whenever available and record every subsequent transformation. For OA carbonate observations, the SDG 14.3.1 metadata template and instructions provide a practical starting structure (UNESCO/IOC, 2019).

At minimum, record where applicable:

→ source repository, DOI or URL, dataset version, licence, and access date;

→ original filename, monitoring programme, and data-provider contact;

- station coordinates and definitions of sites, depth layers, seasons, and observational units;
- variable definitions, original units, final units, and conversion factors;
- sampling gear, sampling effort or volume, sample handling, and analytical method;
- instrument, calibration procedure, reference material, detection limit, and measurement uncertainty;
- quality-flag scheme, original quality flags, mapped flags, and final analysis decisions;
- pH scale, in situ and measurement temperature, and pressure or sampling depth;
- whether each variable was measured, modelled, or derived;
- carbonate-system solver, version, input pair, equilibrium constants, nutrient inputs, and calculation options;
- taxonomic source, accepted identifiers, aggregation level, and name-mapping decisions;
- changes in station location, depth, sampling frequency, gear, instrument, laboratory, method, taxonomy, or processing;
- zero and missing-value rules, exclusions, interpolation, extrapolation, aggregation, and artifact decisions;
- processing-script or workflow version, analyst, processing date, and sufficient information to reproduce the prepared input files.

IV. INPUT DATA FORMATS

Purpose

The objective of this step is to ensure that biotic and abiotic datasets are organized in a standardized, machine-readable format that supports automated processing, documented analytical screening, aggregation, and rate-of-change analysis across multiple sites, biological groups, and abiotic variables.

Implementation

Abiotic data

Abiotic observations should be organized so that each row represents a single observation and each abiotic variable occupies a separate column.

At minimum, abiotic datasets should contain:

- site identifier;
- sampling date (year-month-day);
- depth, pressure, or defined layer identifier, or other observational-unit information where relevant;
- abiotic variables in separate columns.

Biotic data

Biotic observations should be organized so that each row represents a single observation of a biological trait for a specific biological group.

At minimum, biotic datasets should contain:

- site identifier;
- sampling date (year-month-day);
- depth, layer, or other observational-unit information where relevant;
- biological group identifier (bio_group) with listed bio_categories;
- biological trait identifier (bio_trait) with measured or calculated biotic value;

Note: The current automated implementation described in **Section VI** estimates rates of change for trajectories defined by unique combinations of grid_id and bio_group. Consequently, any biological subdivisions that should be analysed separately (e.g. depth layers, seasons, habitats, mesh-size classes, or other observational strata) must be represented as separate trajectories in the input dataset. In practice, this can be achieved by encoding these subdivisions within either the site identifier (grid_id) or the biological group (bio_group) prior to analysis. Abiotic data should be structured consistently so that each biological trajectory is paired with the appropriate abiotic time series.

Convert wide tables to long format

Biotic datasets are frequently stored in wide format, where taxa, traits, or categories occupy separate columns. Although convenient for data storage, this structure is unsuitable for automated time-series analysis.

Whenever possible, biotic data should be converted to long (tidy) format, where:

- each row represents a single observation;
- biological groups are stored in a dedicated column;
- measured values of biological traits are stored in a dedicated column.

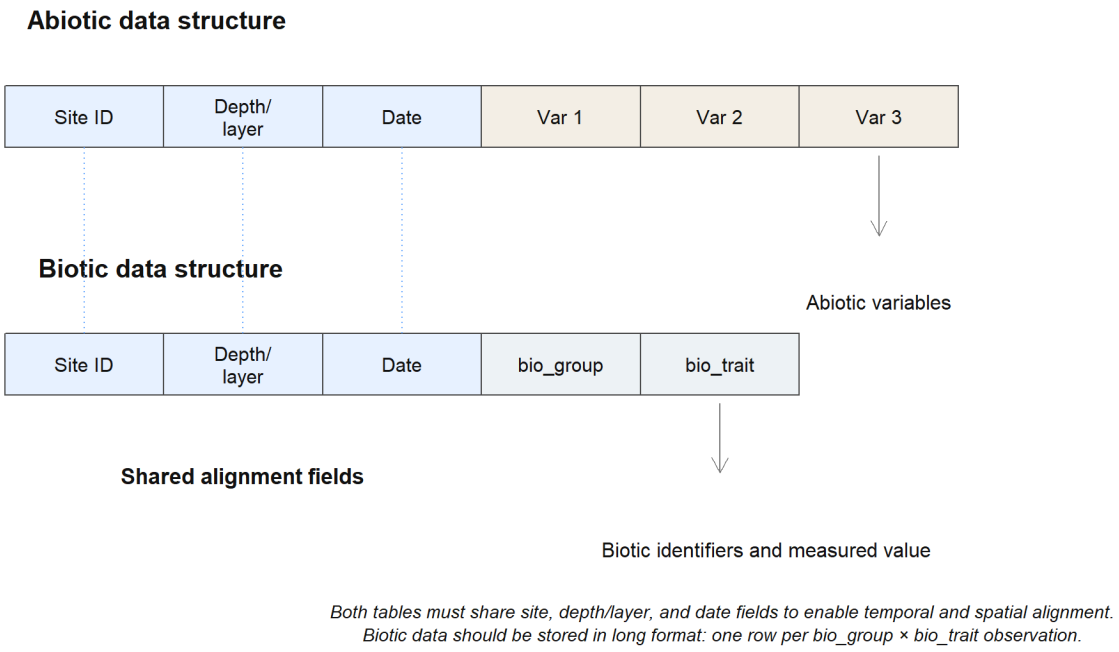


Figure 3. Simplified example of abiotic and biotic table structures used for temporal and spatial alignment.

Abiotic time series should be prepared at the same ecological scale as the biotic observational units. Consequently, a single site may contain multiple abiotic time series representing different seasons, depth layers, or season-layer combinations. Multiple sites may be retained within the same abiotic dataset, provided that station identity remains explicit and abiotic trends can be estimated separately where required.

V. ANALYTICAL WORKFLOW ROC

The rate-of-change (RoC) analytical workflow assumes that biotic and abiotic time series have been prepared according to the procedures described in **Sections II** - **IV**. The workflow analyses biotic and abiotic time series as trajectories of change from which rates of change can be derived.

The following steps are implemented in the automated workflow described in **Section V**, but they can also be performed manually.

1. Abiotic stream: deseason abiotic time series

Abiotic variables often show strong recurring seasonal cycles. These cycles can be larger than the long-term trend and may bias rate estimation if not removed. Therefore, abiotic trends are estimated from deseasoned time series. The workflow follows the deseasonalisation approach described by Sutton et al. (2022), which is appropriate for abiotic variables with strong and stable seasonal structure. The core principles of this approach, as implemented in the automated pipeline, are summarized below. For more details please, see Sutton et al. (2022).

The observed abiotic series can be represented as:

$$v(t) = \alpha + \omega t + \varepsilon(t),$$

where *t* is time expressed as a decimal year, *α* is the intercept coefficient, ω is the long-term linear trend, and *ε(t)* represents residual variability.

Then, the mean residual for each month $\bar{r}_m$ is computed as:

$$r(t) = v(t) - (\alpha + \omega t),$$

$$\bar{r}_m = \frac{1}{n_m} \sum_{t \in m} r(t),$$

where *r(t)* represents the residuals, n_m is the number of observations in month m. The seasonal component is estimated from monthly residuals and removed from the observed series.

$$v_{\mathrm{ds}}(t) = v(t) - \bar{r}_m.$$

The resulting deseasoned series ($v_{ds}(t)$) preserves long-term change while reducing recurring seasonal variability. The adjustment here is applied at the observation level prior to temporal aggregation. Abiotic trends are subsequently estimated from deseasoned monthly means derived from the original observations. The output of this step is a deseasoned abiotic trajectory used for long-term trend estimation.

To assess how rapidly a trend can be distinguished from background variability, Sutton et al. (2022) recommended calculating the Trend Detection Time (TDT) following (Weatherhead et al., 1998). TDT estimates the number of year required for a linear trend to emerge from background variability and autocorrelation:

$$\mathrm{TDT} = \left(\frac{3.3\,\sigma_N}{|\,\omega\,|} \sqrt{\frac{1+\phi}{1-\phi}} \right)^{2/3},$$

where ω - is the estimated linear trend, σ_N is the standard deviation of the residuals, ϕ is the lag-1 autocorrelation of the residuals. In the workflow TDT is reported as a diagnostic metric and does not influence rate-of-change calculations.

2. Segment biological response trajectories

The trajectories of biotic time series can exhibit both linear and non-linear dynamics through time. Biological groups may experience periods of increase, decline, stability, recovery, or transition associated with ecological succession, abiotic change, species interactions, or disturbance events. Because the objective of the framework is to estimate biologically interpretable rates of change, fitting a single linear trend across the entire observation period may obscure important temporal structure.

This approach follows the general logic of segmented straight-line approximation and partition regression, where an ordered trajectory is divided into contiguous intervals within which separate linear models are fitted (Bellman and Roth, 1969; Guthery, 1974). The workflow therefore evaluates whether a biological trajectory is better represented by a single linear trend or by two internally consistent linear segments separated by a breakpoint. The objective is not to maximize model complexity, but to identify biologically meaningful periods of change while avoiding overfitting.

The analysis begins by fitting a full-period linear model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where y_i is the biotic observation, x_i is time, β_0 is the intercept, β_1 is the full-period biotic slope of change, and ε_i is the residual error. This model represents the simplest possible description of biotic change and provides a reference against which segmented alternatives are evaluated. If a biological response trajectory cannot be adequately represented by a single linear trend, a two-segment representation is considered. The trajectory is divided into two contiguous intervals separated by a candidate breakpoint τ:

$$s_1 = \{x_i : x_i \leq \tau\},$$
$$s_2 = \{x_i : x_i > \tau\}.$$

Within each segment, a separate linear regression model is fitted:

$$y_i = \begin{cases} \alpha_1 + \beta_1 x_i + \varepsilon_i, & x_i \le \tau, \\ \alpha_2 + \beta_2 x_i + \varepsilon_i, & x_i > \tau, \end{cases}$$

where α_1 and α_2 are segment-specific intercepts, and β_1 and β_2 are segment-specific slopes.

For each candidate breakpoint, the total residual sum of squares is calculated as:

$$SSE_2(\tau) = \sum_{x_i \le \tau} (y_i - \hat{y}_{1,i})^2 + \sum_{x_i > \tau} (y_i - \hat{y}_{2,i})^2,$$

where $\hat{y}_{1,i}$ and $\hat{y}_{2,i}$ are fitted values from the first and second segment models, respectively.

The optimal breakpoint is defined as the breakpoint argument minimizing the total within-segment residual error (Bellman and Roth, 1969; Guthery, 1974):

$$\tau^* = \arg min_\tau SSE_2(\tau).$$

The workflow currently evaluates only two competing representations:

1. a single full-period linear trend;
2. a two-segment linear trajectory.

The workflow can evaluate segmentation using the Bayesian Information Criterion (BIC). The use of BIC follows the general principle of complexity-penalized change-point selection, where additional breakpoints are accepted only when the improvement in fit justifies the increased model complexity (Yao, 1988).

For each candidate model BIC is calculated as:

$$BIC = n \ln\left(\frac{SSE}{n}\right) + k \ln(n),$$

where *n* is the number of observations, *SSE* is the residual sum of squares, and *k* is the number of estimated parameters. For the single-segment model, *k = 2* (intercept and slope). For the two-segment model, *k = 4* or *k = 5*, depending on whether the breakpoint itself is counted as an additional estimated parameter. In the current implementation, we suggest to count the breakpoint as an additional estimated parameter and therefore k = 5.

Segmentation is accepted when:

$$BIC_{two} < BIC_{single}.$$

Additional segmentation safeguards

Regardless of the selected model-selection method, additional safeguards are applied to avoid biologically implausible segmentations. First, all candidate segments must satisfy a minimum segment-length requirement. This prevents very short segments from being interpreted as independent ecological phases. Second, breakpoints occurring too close to the beginning or end of the trajectory are excluded through a boundary filter. This reduces the risk of creating segments dominated by edge effects. Third, users may optionally require segmentation to achieve a minimum improvement in adjusted R^2 relative to the full-period model. If this criterion is enabled, segmentation is accepted only when the improvement exceeds the user-defined threshold.

If any of these conditions are not satisfied, the biological response trajectory is retained as a single segment.

The output of this step is therefore either:

- one full-period linear rate of biotic change; or
- two biologically interpretable linear segments with different rates of change before and after the identified breakpoint.

3. Estimate biotic rates of change

Biotic rates of change are calculated separately for each accepted biotic segment. For each segment, predicted biotic values are obtained at the beginning and end of the segment from the segment-specific linear model. Because biological traits differ in units and magnitude, biotic rates must be normalized as a dimensionless measure of biotic change ($RC_{s_i}^{bio}(\%)$). This makes rates more comparable across taxa, traits, and biological indicators. To do so, the rate-of-change ($RC_{s_i}^{bio}$) is compared to the maximum predicted value at the segment boundaries. The maximum fitted boundary value is used to avoid inflation of percentage rates when one segment boundary approaches zero.

The normalized biotic rate-of-change is calculated as:

$$\hat{Y}_{start,s_i}^{bio} = a_{s_i} + b_{s_i} x_{1,s_i},$$

$$\hat{Y}_{end,s_i}^{bio} = a_{s_i} + b_{s_i} x_{2,s_i},$$

$$RC_{s_i}^{bio} = \frac{\hat{Y}_{end,s_i}^{bio} - \hat{Y}_{start,s_i}^{bio}}{x_{2,s_i} - x_{1,s_i}},$$

$$RC_{s_i}^{bio}(\%) = 100 \times \frac{RC_{s_i}^{bio}}{\max\left(\hat{Y}_{start,s_i}^{bio}, \hat{Y}_{end,s_i}^{bio}\right)},$$

where:

$\hat{Y}^{bio}_{start,s_i}$ and $\hat{Y}^{bio}_{end,s_i}$ are start and end fitted values of the linear model within a biological response trajectory linear segment s_i;

x_{1,s_i} and x_{2,s_i} are the first and the last year of observation within the linear segment;

$\max(\hat{Y}^{bio}_{start,s_i}, \hat{Y}^{bio}_{end,s_i})$ is a rule to select the maximum of two.

4. Estimate abiotic rates of change

Abiotic rates of change are estimated from the deseasoned abiotic trajectory over the full aligned analysis period. The predicted abiotic values are:

$$\hat{Y}^{env}_{start,S_0} = \alpha + \omega x_{1,S_0},$$
$$\hat{Y}^{env}_{end,S_0} = \alpha + \omega x_{2,S_0},$$

where $\omega$is the full-period abiotic slope. Then the abiotic rate-of-change is calculated as:

$$RC^{env}_{S_0} = \frac{\hat{Y}^{env}_{end,S_0} - \hat{Y}^{env}_{start,S_0}}{x_{2,S_0} - x_{1,S_0}}.$$

Unlike biotic rates, abiotic rates are not normalized because abiotic variables have standardized physical units and their absolute rate is directly interpretable.

5. Pair biotic and abiotic rates

The final step is to pair each biotic rate with the corresponding abiotic rate estimated for the same ecological context and aligned time window.

If the biological response trajectory is represented by one segment, one paired estimate is produced:

$$\left(RC^{bio}_{S_i}(\%), RC^{env}_{S_0}\right).$$

If the biological trajectory is represented by two segments, two paired estimates are produced:

$$\left(RC^{bio}_{S_1}, RC^{env}_{S_0}\right), \left(RC^{bio}_{S_2}, RC^{env}_{S_0}\right).$$

Thus, more than one biotic rate may correspond to the same abiotic rate when a biological trajectory contains multiple segments within the same abiotic analysis period. The final paired dataset forms the basis for biotic-abiotic rate comparison.

VI. PRACTICAL IMPLEMENTATION OF THE AUTOMATED TOOL

1. Important note on data preparation

We strongly recommend completing the data preparation procedures described in **Sections II–IV** before running the automated workflow. Careful data preparation remains essential for ensuring biotic and abiotic time series are ecologically interpretable, methodologically consistent, and suitable for rate-of-change analysis. The tool is not intended to replace expert data preparation, but to provide a standardized implementation of the rate-of-change analysis once biotic and abiotic time series have been prepared, harmonized, and evaluated. Nevertheless, the tool performs a limited number of automated preprocessing steps before rate estimation.

The automated checks do not validate calibration, laboratory procedures, taxonomy, measurement units, pH scale, provider quality flags, undocumented methodological discontinuities, ecological representativeness, or measurement uncertainty. These issues must be resolved and documented before the prepared files are supplied to the workflow.

For the biotic data stream, these include:

1. filtering observations to the selected analysis period;
2. aggregation of multiple observations within a year using user-defined functions (sum, mean, or median);
3. optional completion of missing biological group-year combinations as zeros;
4. screening of biotic time series based on minimum numbers of observed and non-zero years;
5. preparation of the final yearly input dataset used for segmentation.

For the abiotic data stream, these include:

6. conversion and standardization of date formats;
7. filtering observations to the selected analysis period;
8. conversion of selected abiotic variables to numeric format;
9. screening of abiotic variables based on minimum data requirements for trend estimation.

2. Software requirements

The workflow is implemented in the R programming language and is designed to run on Windows, macOS, and Linux operating systems.

Prior to analysis, the following software must be installed:

- R (version 4.2 or later recommended);
- RStudio (optional but recommended);
- Microsoft Excel or compatible spreadsheet software capable of editing .xlsx files.

The RoC workflow requires the R packages *readxl*, *writexl*, *dplyr*, *tidyr*, *lubridate*, *ggplot2*, *tibble*, and *ggrepel*, with *rstudioapi* additionally used when launching it interactively from RStudio. The workflow automatically loads all required R packages during execution. If a package is not installed, the workflow attempts to install it automatically when permissions and internet access are available. Internet access may therefore be required during the first execution.

The tool folder and example time series can be downloaded to user's computer from the open repository OAPoCfNRS/RoC-tool, following the link https://github.com/OAPoCfNRS/RoC-tool.

3. Structure

The analytical framework described above is implemented as a standardized automated workflow in R. The objective of the tool is to provide a reproducible and transparent procedure for transforming biotic and abiotic time series into comparable rates of change while minimizing manual processing and analytical inconsistency among case studies.

The workflow has been designed to operate from a self-contained project folder and does not require manual modification of the analytical code. The workflow is controlled through a single configuration file (settings_configuration.xlsx) and requires only two user-provided inputs: a biotic and an abiotic dataset. The analytical engine automatically performs input-format and internal-consistency checks, selected completeness screening, abiotic preprocessing, biological trajectory segmentation, rate-of-change estimation, and generation of paired biotic-abiotic rate datasets.

The workflow expects a standardized directory structure organized within a folder *RoC/* which consists of five components (**Figure 4**):

- *data/* – a folder with input datasets and metadata;
- *settings_configuration.xlsx* – user-defined analysis settings;

- *RUN.R* – workflow launcher;
- *DO_NOT_RUN.R* – analytical engine containing all computational procedures.
- README.md - contains basic instructions about how to launch the analytical workflow.

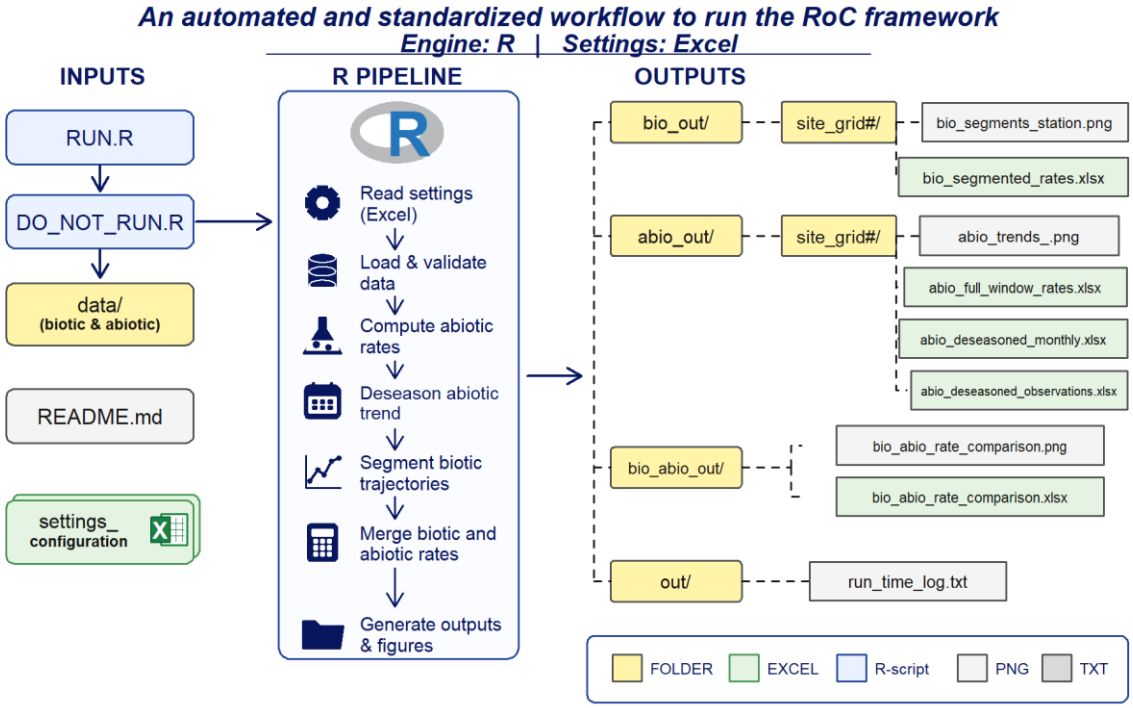


Figure 4. Structure of the automated workflow.

Users interact only with the input datasets and the settings file. All analytical calculations are executed automatically by the workflow in R.

The data/ directory is where the biotic and abiotic datasets will need to be copied in. The analytical engine automatically reads all required inputs using the file names specified in the configuration file.

The file RUN.R acts as the workflow launcher and should be the only script executed directly by the analyst.

The file DO_NOT_RUN.R contains the complete analytical implementation and workflow does not require manual modification of analytical code.

4. Preparing input files

Prior to analysis, biotic and abiotic time series must be prepared according to the procedures described in **Sections II-IV**. The workflow assumes that measurement QA/QC, provider-flag interpretation, harmonization, analytical completeness screening, temporal standardization,

seasonal decisions, observational-unit definitions, and companion metadata have already been completed.

Abiotic observations should be stored in a single table containing sampling date, site identifier, depth or layer information, and abiotic variables in separate columns. Biotic observations should be stored in long format, where each row represents a single biotic observation (bio_unit) associated with a specific biological group (bio_group) and biological trait (bio_trait). Abiotic and biotic observations do not need to share identical temporal resolution. However, they must represent the same ecological system and overlap sufficiently in time to permit paired rate estimation.

Demonstration dataset

For demonstration purposes, the *data/* folder already contains example biotic and abiotic time series derived from Copernicus Marine Service reanalysis products, as daily surface data extracted at approximately 0.5-2 m depth. The dataset includes nine surface sites distributed across the Iberian-Biscay-Irish (IBI), Baltic Sea, and Mediterranean Sea regions.

The abiotic time series include:

- sea-water potential temperature (T);
- salinity (S);
- pH;
- partial pressure of carbon dioxide (pCO_2);
- additional carbonate-system variables derived from carbonate chemistry calculations.

The biotic time series is represented by net primary production (nppv), which serves as a proxy for phytoplankton primary production.

Table 3. Demonstration datasets used in the guide.

Variables	**Copernicus product**	**Dataset**
Temperature (T), Salinity (S)	IBI_MULTIYEAR_PHY_005_002 , (European Union-Copernicus Marine Service, 2020a)	cmems_mod_ibi_phy-temp_my_0.027deg_P1D-m
Temperature (T), Salinity (S)	BALTICSEA_MULTIYEAR_PHY_003_011, (European Union-Copernicus Marine Service, 2018a)	cmems_mod_bal_phy_my_P1D-m

Temperature (T), Salinity (S)	MEDSEA_MULTIYEAR_PHY_006_004, (European Union-Copernicus Marine Service, 2025a)	cmems_mod_med_phy-temp_my_4.2km_P1D-m
pH, pCO_2	IBI_MULTIYEAR_BGC_005_003 (European Union-Copernicus Marine Service, 2020b)	cmems_mod_ibi_bgc-car_my_0.027deg_P1D-m
pH, pCO_2	BALTICSEA_MULTIYEAR_BGC_003_012, (European Union-Copernicus Marine Service, 2018b)	cmems_mod_bal_bgc_my_P1D-m
pH, pCO_2	MEDSEA_MULTIYEAR_BGC_006_008, (European Union-Copernicus Marine Service, 2025b)	cmems_mod_med_bgc-co2_my_4.2km_P1D-m
Net primary production (nppv)	IBI_MULTIYEAR_BGC_005_003, (European Union-Copernicus Marine Service, 2020b)	cmems_mod_ibi_bgc-plankton_my_0.027deg_P1D-m
Net primary production (nppv)	BALTICSEA_MULTIYEAR_BGC_003_012, (European Union-Copernicus Marine Service, 2018b)	cmems_mod_bal_bgc_my_P1D-m
Net primary production (nppv)	MEDSEA_MULTIYEAR_BGC_006_008, (European Union-Copernicus Marine Service, 2025b)	cmems_mod_med_bgc-bio_my_4.2km_P1D-m

Table 4. Demonstration sites used in the guide.

Site ID	Region	Longitude	Latitude	Approx. extraction depth (m)
IBI_1	Iberian–Biscay–Irish	-10.00587	42.50571	1.54
IBI_2	Iberian–Biscay–Irish	-4.50042	50.25041	1.54
IBI_3	Iberian–Biscay–Irish	-6.00190	46.00159	1.54
BAL_1	Baltic Sea	15.98045	55.24994	1.52
BAL_2	Baltic Sea	12.75008	55.74991	1.52
BAL_3	Baltic Sea	20.05580	57.31852	1.52
MED_1	Mediterranean Sea	12.80088	44.80029	1.02
MED_2	Mediterranean Sea	25.19958	38.99991	1.02
MED_3	Mediterranean Sea	5.50025	42.79969	1.02

Additional carbonate chemistry variables were calculated in R using the *seacarb* package (Gattuso et al., 2024) from pH, pCO_2, temperature, and salinity, using the default equilibrium constants implemented in seacarb. Derived variables include dissolved inorganic carbon (DIC), total alkalinity (TA), bicarbonate ion concentration (HCO_3^-), carbonate ion concentration (CO_3^{2-}), aragonite saturation state (Ωarag), and calcite saturation state (Ωcalc).

These calculated variables are included for workflow demonstration and are explicitly treated as derived rather than measured quantities. The pH + pCO2 input pair does not constrain all derived carbonate variables equally well. Applications using these outputs for scientific interpretation should document the pH scale, solver settings, input uncertainties, and propagated uncertainty,

and should not assume that derived DIC or TA has the same measurement quality as directly observed DIC or TA (UNESCO/IOC, 2019; Orr et al., 2018).

5. Configure the analysis

All user-defined options are controlled through the Excel configuration file *settings_configuration.xlsx*. The R scripts do not need to be edited for routine analyses. The settings file contains several worksheets, each controlling a specific part of the workflow. Sheet names and required keys should not be changed.

The configuration file contains separate worksheets controlling:

- project settings;
- biotic preprocessing;
- abiotic segmentation;
- abiotic preprocessing;
- abiotic variable selection;
- biotic column mapping;
- abiotic column mapping.

Readme

The *Readme* worksheet contains simple general rules of configuration. Please read the worksheet before proceeding (**Figure 5**).

1	**Settings file for the BIO + CHEM rate-of-change pipeline**
2	
3	How to use this file
4	1. Project: set input file names, output folder, and whether to run BIO, ABIO, or both.
5	2. Bio_ColMap: map biological data columns from your file to the required standard names.
6	3. Bio_settings: choose biological analysis years and yearly aggregation rules.
7	4. Bio_segmentation: control biological segmentation.
8	5. Abio_ColMap: map only structural abiotic columns such as date and site ID.
9	6. Abio_selected_variables: paste any abiotic variables you want to analyse. No predefined variables are required.
10	7. Abio_settings: choose abiotic analysis years and trend/QC thresholds.
11	
12	Core pipeline logic
13	BIO: aggregate to one value per year per biological group and site, then fit one or two linear segments.
14	ABIO: deseason each selected variable, aggregate to monthly means, then estimate one full-window trend.
15	BIO x ABIO: each biological segment is paired with the same full-window environmental rate for the matching site and selected variable.
16	
17	Important rules
18	Column names in Abio_selected_variables must exactly match the column names in the abiotic input file after Abio_ColMap is applied.
19	Column maps are for renaming columns only. Do not list selected ABIO variables in Abio_ColMap.
20	Keep sheet names unchanged.
21	Do not delete required keys; change only values in the yellow cells.
22	
23	RoC v5 segmentation note
24	Bio_segmentation includes BIC model-selection method.
25	When selection_method = bic, min_segment_years and boundary_fraction, apply_min_r2_gain_filter = TRUE, min_r2_gain is also required before accepting two segments.

Figure 5. Readme worksheet in the *settings_configuration.xlsx* configuration file.

Project settings

Open the worksheet *Project* (**Figure 6**). This sheet defines the project folder, input files, output folder, and which parts of the workflow should be executed.

The main fields are:

project_dir – folder containing the launcher and settings file. Use "." when the settings file is located in the main project folder.

data_dir – folder containing input datasets. By default this is data.

out_dir – folder where outputs will be saved. By default this is out.

bio_infile – name of the biotic input file located inside data_dir.

abio _infile – name of the abiotic input file located inside data_dir.

run_bio – set to TRUE to run the biotic pipeline.

run_abio – set to TRUE to run the abiotic pipeline.

overwrite_outputs – set to TRUE if existing outputs may be overwritten.

log_runs – set to TRUE to write a run log.

For a full biotic-abiotic rate comparison, both *run_bio* and *run_abio* should be enabled. If only biotic segmentation or only abiotic trend estimation is required, one pipeline may be disabled.

Example:

For the provided example, biotic time series stored in *data/* folder are titled *biotic.xlsx*, and the abiotic time series are titled *abiotic.xlsx*.

key	value	description
project_dir	.	Folder containing this settings file and launcher. Use . for current folder.
data_dir	data	Folder with input data files. Relative paths are resolved from project_dir.
out_dir	out	Main output folder.
bio_infile	biotic.xlsx	Biotic input file name inside data_dir.
abio_infile	abiotic.xlsx	Abiotic input file name inside data_dir.
run_bio	TRUE	TRUE/FALSE. Run biotic pipeline with processing and segmentation.
run_abio	TRUE	TRUE/FALSE. Run abiotic processing.
overwrite_outputs	TRUE	TRUE/FALSE. Allow overwriting existing output files.
log_runs	TRUE	TRUE/FALSE. Write a simple run log.

Figure 6. *Project* worksheet in the settings.xlsx configuration file.

Map biotic input columns

Open the worksheet *Bio_ColMap*. This sheet tells the workflow how columns in the biotic input file correspond to the standardized column names required by the tool.

The required operated column names are:

grid_id – site, station, grid cell, or location identifier;

year – sampling year;

bio_group – biological subject of assessment, such as a taxon, functional group, size class, or ecosystem process variable;

bio_trait_var – biological response value.

In the original column, enter the column names exactly as they appear in the biotic input file. In the operated column, retain the standardized names expected by the workflow.

Example:

For the provided example, NPPV can be treated as a biological proxy of primary production. The biotic input should therefore contain one column identifying the biological group, for example nppv in *bio_group* column, and one numeric column containing the nppv value in bio_trait_var column, as shown in **Figure 7**.

	A	B	C	D
1	**original**	**operated**	**required?**	**description**
2	station	grid_id	yes	Site, station, grid, or location identifier.
3	year	year	yes	Sampling year. If your biological data uses full dates, map/derive year before running or extend the pipeline.
4	bio_group	bio_group	yes	Biological unit: taxon, genus, species, size class, functional group, etc.
5	nppv_value	bio_trait_var	yes	Biological response value: count, biomass, cover, growth rate, etc.

Figure 7. *Bio_ColMap* worksheet in the settings.xlsx configuration file.

Configure biotic settings

Open the worksheet *Bio_settings*. This sheet controls the biotic analysis window, aggregation method, and basic screening rules (**Figure 8**).

The main fields are:

analysis_years – biotic analysis window written as start;end, for example 2000;2024;

aggregation – method used to combine multiple biotic rows within the same grid_id × year × bio_group; allowed values are sum, mean, and median;

complete_missing_groups_as_zero – controls whether missing biological group-year combinations are completed as zeros;

min_non_missing_years – minimum number of non-missing years required for a biological group to be analysed;

min_non_zero_years – optional filter requiring a minimum number of years with values greater than zero.

The choice of aggregation should follow the type of the biological organization level. Abundance-type data may often use sum, while biomass, productivity, or concentration-like variables often require mean or median. Aggregation combines multiple biotic observations occurring within the same grid_id × year × bio_group combination into a single annual value used for rate estimation.

The setting complete_missing_groups_as_zero should only be set to TRUE when missing group–year combinations represent observed absence. It is another extra helper. For biotic measurements where zeros or omissions do not represent true absence, this setting should be set to FALSE.

The settings *min_non_missing_years* and *min_non_zero_years* are optional filters which can be selected based on the expert knowledge.

Example:

For the Copernicus example, the time series represent the time window between 2000 and 2024 (25 years in total). And we set that in case of duplicated bio_unit – the values must be averaged, as they are probably indicate the same observation (although in fact they are not expected from the reanalysis output). We do not want to substitute the missing values with zero, as it is a modelled data with the full cover. The thresholds for *min_non_missing_years* and *min_non_zero_years* should be selected based on data availability, time-series length, and the objectives of the analysis.

key	value	description
analysis_years	2000;2024	BIO analysis window as start;end. Only these years are used for biological segmentation and rates.
aggregation	mean	How to combine multiple BIO rows within the same grid_id + year + bio_group. Allowed: sum, mean, median.
complete_missing_groups_as_zero	FALSE	TRUE means missing group-year combinations are treated as observed zero values. FALSE keeps only observed rows.
min_non_missing_years	3	Minimum number of non-missing years required for a BIO group to be analysed.
min_non_zero_years	3	Optional filter. Minimum years with value > 0. Use 0 to keep true-zero dominated series.

Figure 8. *Bio_settings* worksheet in the settings.xlsx configuration file.

Configure biotic segmentation

Open the worksheet *Bio_segmentation*. This sheet controls whether biological response trajectories may be represented by one or two linear segments (**Figure 9**).

The main fields are:

enabled – set to TRUE to allow segmentation; set to FALSE to force one full-period biotic trend;

min_segment_years – minimum number of observations or years required in each biotic segment;

max_segments – maximum number of biotic segments; the current workflow supports one or two segments;

min_r2_gain – minimum adjusted R^2 improvement required to accept a two-segment model;

boundary_fraction – protects the beginning and end of the time series from breakpoints too close to the boundaries.

selection_method – model-selection approach used to evaluate segmentation. BIC is the only supported option.

bic_count_breakpoint – if TRUE, the breakpoint is counted as an additional parameter in the BIC calculation (normally TRUE).

apply_min_r2_gain_filter – if TRUE, segmentation must additionally satisfy the minimum adjusted-R^2 improvement criterion.

The min_segment_years can be selected arbitrarily. It can be suggested to start with 1/3 of the total number of years in the global time window.

The minimum adjusted R^2 threshold can be selected based on the length and variability of the time series. The filter may also be disabled entirely by setting *apply_min_r2_gain_filter* = FALSE, regardless of the selected segmentation method. The default value of *min_r2_gain* is 0.05.

The boundary fraction is additional parameter to the *min_segment_years,* only is expressed as a fraction of time series close to the edges where the breakpoint is not allowed. Users may adjust either the minimum number of observations per segment (min_segment_years) or the boundary exclusion zone (boundary_fraction) to control segmentation conservatism. However, candidate breakpoints must satisfy both criteria simultaneously. For example, if the series length is 25 years, then 25% it is only 6.5 years. The *min_segment_years* set to 8 will not allow this scenario.

More conservative settings reduce segmentation. For example, increasing *min_r2_gain*, increasing *min_segment_years* / increasing *boundary_fraction* will make it harder for the workflow to accept a two-segment biological trajectory.

For exploratory analyses, segmentation can be enabled and diagnostics inspected. For conservative rate estimation, the segmentation settings should be chosen before interpreting outputs and not adjusted repeatedly to obtain a preferred pattern.

key	value	description
enabled	TRUE	TRUE = allow one/two-segment biological representation. FALSE = force one full-period BIO line.
min_segment_years	8	Minimum number of observations/years per biological segment.
max_segments	2	Maximum number of BIO segments. Keep 2 to match the current method.
min_r2_gain	0.1	Minimum adjusted R^2 improvement required when apply_min_r2_gain_filter = TRUE. Use 0 to disable by threshold.
boundary_fraction	0.25	Prevents breakpoints too close to the start/end. Example 0.25 = outer 25% protected.
selection_method	bic	Segmentation model-selection method. Allowed: bic.
bic_count_breakpoint	TRUE	TRUE recommended. Counts the estimated breakpoint as an additional BIC parameter: one-line k=2; two-segment k=5. FALSE uses k=4 for two segments.
apply_min_r2_gain_filter	TRUE	TRUE applies the min_r2_gain filter after the selected method. Keep TRUE for BIC + additional ecological filter; set FALSE for pure BIC after length/boundary filters.

Figure 9. *Bio_segmentation* worksheet in the settings.xlsx configuration file.

Map abiotic input columns

Open the worksheet Abio_*ColMap*. This sheet maps only the structural environmental columns required by the workflow (Figure 10).

The required operated column names are:

date – full sampling date;

grid_id – site, station, grid cell, or location identifier.

The *grid_id* values in the abiotic file must match the *grid_id* values in the biotic file if biotic and abiotic rates are to be paired. Multiple stations may be stored in the same abiotic file, but station identity must remain explicit.

Abiotic variables themselves should not be listed in *Abio_ColMap*. Variables selected for trend analysis are defined separately in *Abio_selected_variables*.

original	operated	required?	description
Date	date	yes	Full sampling date. Required for deseasoning and monthly aggregation.
Station	grid_id	yes	Site, station, grid, or location identifier. Must match BIO grid_id for BIO x ABIO pairing.

Figure 10. *Abio_ColMap* worksheet in the settings.xlsx configuration file.

Select abiotic variables

Open the worksheet *Abio_selected_variables*. This sheet defines which abiotic variables will be analysed (**Figure 11**).

For each variable:

- set *enabled* to TRUE if the variable should be analysed;
- enter the exact variable name in 'variable' as it is in the environmental time series;
- in C column provide a readable title to be reflected in the output tables and plots;
- in D column provide an output plot label in *ylab*.

Variables listed with enabled = FALSE remain in the settings file but are not analysed.

enabled	variable	title	ylab	description
TRUE	T	Temperature	Temperature	Paste exact variable name from ABIO data after Abio_ColMap. Example: oxygen, salinity, heatwave_days, aragonite_saturation.
TRUE	pH	pH	pH	Optional title/ylab are used only in plots and tables. Leave blank to use variable name.
TRUE	H+	H+ proton	H+ proton	Set enabled=FALSE to keep a variable listed but not analyse it.
TRUE	S	Salinity	Salinity	
TRUE	ALK	Total alkalinity	Total alkalinity	
TRUE	pCO2	Partial pressure of CO2	pCO2	
TRUE	OmegaAragonite	Aragonite saturation	Ω Arag	
TRUE	OmegaCalcite	Calcite saturation	Ω Calc	
FALSE	no3	Nitrates	NO3	
FALSE	si	Silicates	Si	
FALSE	po4	Phosphates	PO4	
FALSE				

Figure 11. *Abio_selected_variables* worksheet in the settings.xlsx configuration file.

Configure abiotic settings

Open the worksheet *Abio_settings* (**Figure 12**). This sheet defines the abiotic analysis window and minimum data requirements for abiotic trend estimation.

The main fields are:

analysis_years – abiotic analysis window written as start;end, for example 2000;2024;

min_months_for_trend – minimum number of deseasoned monthly observations required to estimate an abiotic trend;

tdt_min_months – minimum number of monthly observations required.

The abiotic analysis window should overlap with the biotic analysis window. It does not need to contain observations at the same dates as the biotic series, but it must represent the same ecological context.

key	value	description
analysis_years	2000;2024	Abiotic analysis window as start;end. Full-window trend and rate are calculated over this period.
min_months_for_trend	24	Minimum deseasoned monthly observations required for ABIO trend estimation.
tdt_min_months	60	Minimum monthly observations for OK Time of Detection QC. Shorter series may still run but are flagged.

Figure 12. *Abio_settings* worksheet in the settings.xlsx configuration file.

Check configuration consistency before running

Before running the workflow, check that:

- all input files listed in *Project* exist inside the data folder;
- all required biotic columns are mapped in *Bio_ColMap*;
- all required abiotic structural columns are mapped in *Abio_ColMap*;
- *grid_id* values are consistent between biotic and abiotic files;
- biotic and abiotic analysis windows are appropriate;
- selected abiotic variables exist in the abiotic input file;
- segmentation settings are appropriate for the length and structure of the biotic time series;
- *complete_missing_groups_as_zero* matches the biological meaning of missing group–year combinations.

6. Run the workflow

Launching the workflow

After the input datasets have been prepared and the configuration file has been completed, the workflow can be executed by running RUN.R (**Figure 13**). Open the *ROC/* folder. To start the analysis, open RUN.R in RStudio (double-click the file) and execute the script by pressing Ctrl+Shift+Enter, or by selecting all lines (Ctrl+A) and pressing the button 'Run' in the upper-right corner of the script panel. This is the only script that should be run directly by users.

During execution, the launcher automatically:

1. identifies the project directory;
2. sets the output directory;

3. checks that the configuration file is present;
4. checks that the analytical engine is present;
5. loads all workflow settings;
6. validates the input datasets;
7. launches the analytical pipeline.

No manual execution of individual functions is required.

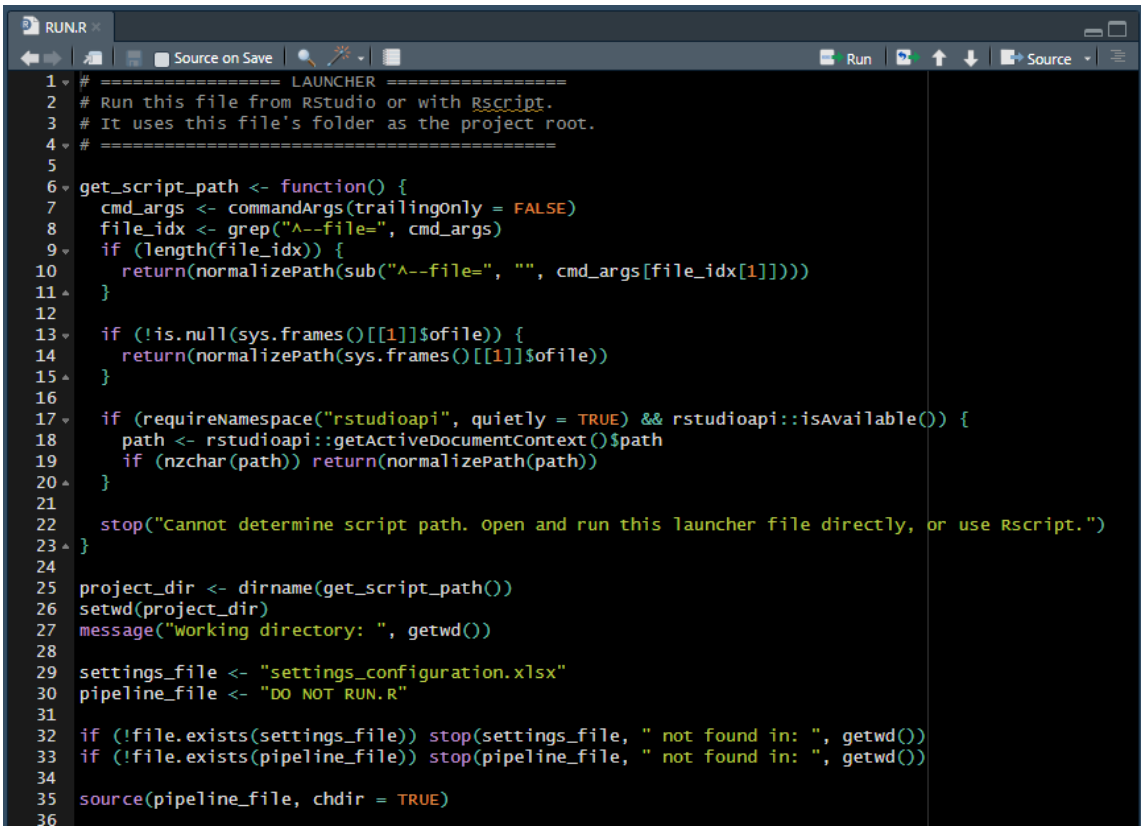


Figure 13. RUN.R script launcher opened in R.

Monitoring progress

As the workflow runs, progress messages are displayed in the R console (**Figure 14**). Depending on dataset size, execution may take from a few seconds to several minutes. The first execution may take longer than subsequent runs because required R packages may need to be installed automatically.

Under normal conditions, the workflow should finish without errors and return a completion message in the console. If an error occurs, the message usually identifies the specific configuration setting, missing file, or input column responsible for the interruption.

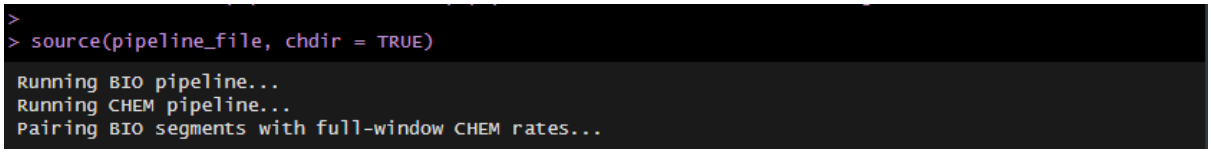


Figure 14. The analytical workflow progress shown in R console.

Verifying successful execution

A successful run generates a set of output directories automatically. Depending on the selected workflow options, the project folder will contain output folders similar to:

out/

- → bio/
  - → biological trajectory diagnostics;
  - → biotic rate estimates;
  - → abio/abiotic trend estimates;
  - → deseasoned abiotic series;
- → bio_abio/
- → workflow logs.

The file *run.log* records workflow execution details and can be used for troubleshooting or reproducibility purposes. Each time the workflow is executed, output files may be overwritten if the *overwrite_outputs* is set TRUE in the configuration file. Users should therefore archive previous results before re-running the workflow if multiple versions of the analysis need to be retained.

Before interpreting results, inspect the generated diagnostic figures and summary tables to verify that biotic segmentation, abiotic trend estimation, and observational-unit definitions were applied as intended.

7. Understanding the outputs

The workflow generates several output folders containing intermediate products, quality-control diagnostics, visualizations, and final rate estimates. The exact contents depend on the selected workflow options and input data structure, but the outputs can be grouped into three categories:

1. biotic outputs (*bio/*);
2. abiotic outputs (*abio/*);
3. paired biotic-abiotic outputs (*bio_abio/*).

Biotic outputs

The biotic output directory contains the processed biological trajectories, segmentation results, diagnostic figures, and biotic rates of change.

The most important outputs are:

1. Biological response trajectory plots

Saved in directory *ROC/out/bio/bio_segments_SITE_p#.png*, where 'SITE reflects the actual station as in the input time series, and 'p#' reflects the page number containing figures for the specific station. The figures contain visualizations of processed biological trajectories with applied linear segmentation when applicable.

2. Biotic time series screening results

Saved in directory *ROC/out/bio/bio_group_screening.xlsx*. Contains information about a number of observed years, non-missing and non-zero years in each *grid_id* (site) specific time series, as well as filtering decision on whether to keep or skip the time series.

3. Biotic rates of change tables

Saved in directory *ROC/out/bio/bio_segmented_rates.xlsx* (**Table 5**). The biotic rates of change and supporting segmentation statistics for each analysed biological trajectory.

The segmentation diagnostics should be inspected before interpreting rates of change. Acceptance of segmentation indicates that the trajectory satisfied the selected model-selection criterion (BIC) and all additional filtering requirements. However, statistical acceptance alone does not guarantee ecological relevance. The fitted segments should therefore be evaluated visually to confirm that the identified breakpoint reflects a meaningful change in biological dynamics rather than isolated observations or residual noise. The objective is to confirm that the

selected segmentation reflects meaningful biological structure rather than isolated observations or noise. If the fitted segmentation does not appear biologically meaningful, the analyst should re-evaluate both the segmentation settings and the suitability of segmentation for that particular trajectory.

Positive *bio_rate_pct_per_year* values indicate increasing biological trajectories, while negative values indicate declining trajectories. Because biotic rates are normalized, values are expressed as relative percentage change and can be compared among biological groups with different units and magnitudes.

Table 5. Output variables saved in *bio_segmented_rates.xlsx*.

Column	**Description**
grid_id	Site, station, or spatial unit identifier.
bio_group	Biological group being analysed (taxon, functional group, biodiversity metric, etc.).
seg_id	Segment identifier within a biological trajectory.
segment_model	Selected model representation (single segment or segmented trajectory).
x1	First year of the segment.
x2	Last year of the segment.
n	Number of observations included in the segment.
intercept	Estimated intercept of the segment-specific linear model.
slope	Estimated slope of the segment-specific linear model.
slope_se	Standard error of the slope estimate.
slope_p	P-value associated with the slope estimate.
r2	Coefficient of determination (R^2) of the fitted segment.
adj_r2	Adjusted R^2 accounting for model complexity.
sse	Sum of squared residual errors for the fitted segment.
breakpoint_year	Year separating adjacent segments when segmentation is accepted.
segmentation_accepted	Indicates whether segmentation was accepted (TRUE) or rejected (FALSE).
segmentation_reason	Reason for accepting or rejecting segmentation.
selection_method	Segmentation method used for model selection (bic).

selection_score_single	Selection score of the single-segment model.
selection_score_two	Selection score of the accepted two-segment model.
bic_single	BIC score of the single-segment model.
bic_two	BIC score of the two-segment model.
bic_delta	Difference between the two-segment and single-segment BIC values. Negative values favour segmentation
r2_gain	Improvement in adjusted R^2 obtained by segmentation relative to the full-period model.
bio_y_start_hat	Predicted biotic value at the beginning of the segment.
bio_y_end_hat	Predicted ~~botic~~biotic value at the end of the segment.
bio_duration_years	Segment duration in years.
bio_abs_change	Total absolute change across the segment.
bio_rate_abs_per_year	Absolute rate-of-change per year.
bio_boundary_max	Larger of the predicted start and end values used for normalization.
bio_rate_pct_per_year	Normalized biotic rate-of-change (% yr^{-1}). Primary biological response metric used in the workflow.
bio_total_pct_change	Total normalized percentage change across the segment.

4. Selected biotic time series for the input

Saved in directory *ROC/out/bio/bio_yearly_input_for_segmentation.xlsx*. The table represents the actual biotic time series analysed by the workflow and therefore provides a transparent record of the data entering the segmentation procedure.

5. Aggregated biotic observations

Saved in directory *ROC/out/bio/bio_yearly_observed_aggregated.xlsx*. This file contains the biotic observations after temporal aggregation to annual values, but before biotic time-series completion, screening, and segmentation.

Abiotic outputs

The abiotic output directory contains the processed abiotic time series, deseasonalisation diagnostics, trend analyses, trend-detection statistics, and abiotic rates of change.

The most important outputs are:

1. Abiotic trajectory plots

Saved in directory *ROC/out/abio/* with standardized names *abio_trends_SITE.png* corresponding to the analysed abiotic site. The figures contain visualizations of the original abiotic observations, monthly means, deseasoned abiotic series, fitted long-term trends, and Trend Detection Time indication. These plots should be inspected to verify that the deseasonalisation procedure successfully removed recurring seasonal variability while preserving long-term abiotic change.

Particular attention should be paid to:

- adequacy of temporal coverage;
- strength of seasonal variability;
- plausibility of the fitted trend;
- potential outliers or artifacts remaining in the time series.

2. Best-fit environmental curve diagnostics

Saved in directory *ROC/out/abio/abio_best_fit_diagnostic_models.xlsx*. This table summarizes statistics for the best-fitting model selected for each abiotic variable and site.

Table 6. Output variables listed in *chem_best_fit_diagnostic_models.xlsx*.

Column	**Description**
grid_id	Site, station, or spatial unit identifier.
abio_var	Abiotic variable analysed.
abio_title	Variable name used in plots and outputs.
best_fit_model	Selected model providing the best fit to the abiotic trajectory.
best_fit_adj_r2	Adjusted R^2 of the selected model.

3. Best-fit model prediction curves

Saved in directory *ROC/out/abio/abio_best_fit_diagnostic_curves.xlsx*. This table contains predicted values from the selected best-fit model and can be used to reconstruct the fitted abiotic trajectory. Candidate models include linear, polynomial, logarithmic, and exponential representations evaluated using adjusted R^2. The fitted curve is not used for the rate-of-change calculation. It is a helper primarily intended for the visual diagnostic evaluation of abiotic time-series structure.

Table 7. Output variables listed in *abio_best_fit_diagnostic_curves.xlsx*.

Column	**Description**
dec_year	Decimal year used for model fitting.
best_fit_value	Predicted abiotic value from the selected model.
best_fit_model	Selected best-fit model.
grid_id	Site, station, or spatial unit identifier.

abio_var	Abiotic variable analysed.

4. Monthly climatology tables

Saved in directory *ROC/out/abio/* with standardized names *abio_deseasoned_monthly.xlsx* corresponding to the analysed abiotic variables. This table contains monthly aggregated abiotic observations before and after deseasonalisation. Monthly climatologies are used internally during the deseasonalisation procedure and are reflected through the relationship between *raw_monthly_mean* and *deseasoned_monthly_mean*.

Table 8. Output variables listed in *chem_deseasoned_monthly.xlsx*.

Column	**Description**
abio_var	Abiotic variable analysed.
year	Calendar year.
dec_year	Decimal year.
grid_id	Site, station, or spatial unit identifier.
monthly_date	Representative monthly date.
raw_monthly_mean	Observed monthly mean before deseasonalisation.
deseasoned_monthly_mean	Monthly mean after removal of recurring seasonal variability.
n_obs	Number of observations contributing to the monthly mean.

5. Deseasoned abiotic observations

Saved in directory *ROC/out/abio/abio_deseasoned_observations.xlsx*. This table contains the observation-level abiotic dataset after data preparation step.

In addition to the original abiotic observations, the table includes:

- monthly climatological residuals;
- deseasoned abiotic variables.
- variables with suffix *_monthly_residual_clim*, representing monthly climatological corrections.
- variables with suffix *_deseasoned,* representing the final deseasoned observations used by the workflow.

6. Abiotic rates of change tables

Saved in directory ROC/out/abio/*abio_full_window_rates.xlsx*. This table contains the final abiotic rates of change calculated from the deseasoned abiotic time series, and Trend Detection

Time (TDT) estimates calculated for each abiotic variable. The TDT estimate should be interpreted cautiously when tdt_qc = FALSE, indicating that the available data length or residual autocorrelation falls outside the recommended range for reliable TDT estimation. The metric *env_rate_per_year* represents the estimated abiotic rate-of-change in physical units per year, and is used for the further workflow steps.

Table 9. Output variables in *chem_full_window_rates.xlsx*.

Column	**Description**
grid_id	Site, station, or spatial unit identifier.
abio_var	Abiotic variable analysed.
abio_title	Variable name used in plots and outputs.
abio_ylab	Plot label used in visualizations.
abio_n_months	Number of monthly observations used in trend estimation.
abio_analysis_start	First year included in the analysis.
abio_analysis_end	Last year included in the analysis.
abio_y_start_hat	Predicted abiotic value at the beginning of the analysis period.
abio_y_end_hat	Predicted abiotic value at the end of the analysis period.
abio_abs_change	Total abiotic change across the analysis period.
abio_rate_abs_per_year	Abiotic rate-of-change per year. Primary abiotic response metric used in the workflow.
abio_slope_se	Standard error of the trend estimate.
abio_slope_p	P-value associated with the trend estimate.
abio_r2	Coefficient of determination (R^2).
abio_adj_r2	Adjusted R^2.
abio_status	Status of abiotic trend estimation.
TDT	Trend Detection Time estimate.
uTDT	Uncertainty associated with TDT.
omega0	Initial trend magnitude used in TDT calculations.
sigmaN	Abiotic noise estimate.
phi	Temporal autocorrelation coefficient.
tdt_n	Number of observations used in TDT estimation.
tdt_qc	TDT quality-control flag.
tdt_qc_reason	Explanation of the TDT quality-control outcome.

Paired biotic-abiotic outputs

The paired biotic-abiotic output directory contains the final products of the workflow. These outputs combine biotic rates of change and abiotic rates of change into a common analytical framework suitable for subsequent ecological interpretation and comparison among sites, biological groups, seasons, and depth layers.

The most important outputs are:

1. Biotic-abiotic rate comparison plots

Saved in directory *ROC/out/bio_abio/bio_abio_rate_comparison.png*. The figures provide a graphical summary of paired biotic and abiotic rates of change. They are intended as exploratory visualization tools and facilitate rapid comparison of biological responses across abiotic gradients, sites, seasons, or biological groups. These plots should be interpreted together with the corresponding rate tables and biotic segmentation diagnostics.

2. Biotic-abiotic rate comparison table

Saved in directory *ROC/out/bio_abio/bio_abio_rate_comparison.xlsx*. This table represents the final paired biotic-abiotic dataset and should be considered the principal output of the workflow for subsequent ecological interpretation, comparison among time-series units, and synthesis across sites and case studies.

Each row of the table represents one biotic segment paired with one abiotic trend for the same biological trajectory (ts_unit). The biotic rate (*bio_rate_pct_per_year*) is calculated for a specific segment of a biological trajectory, whereas the abiotic rate (*chem_rate_abs_per_year*) is calculated for the complete abiotic time series associated with the same observational context. A single time-series unit may contain one or more segments. Consequently, a single time-series unit may generate multiple biotic rates of change. Each biotic segment is paired separately with the corresponding abiotic rate.

The table combines:

- biotic segmentation results from *bio_segmented_rates.xlsx*;
- abiotic trend estimates from *chem_full_window_rates.xlsx*.

Table 10. Output variables in *bio_chem_rate_comparison.xlsx*.

Column	**Description**

point_id	Unique identifier of one biological group-segment. The same point_id is repeated across abiotic variables.
pair_note	Note explaining the pairing logic.
bio_analysis_start	First year of the biotic analysis window.
bio_analysis_end	Last year of the biotic analysis window.
grid_id	Site, station, or spatial unit identifier.
bio_group	Biological group analysed.
seg_id	Segment identifier within the biological trajectory.
segment_model	Selected biotic model: single segment or two-segment trajectory.
x1	First year of the biotic segment.
x2	Last year of the biotic segment.
n	Number of observations in the biotic segment.
intercept	Intercept of the segment-specific biotic linear model.
slope	Slope of the segment-specific biotic linear model.
slope_se	Standard error of the biotic slope.
slope_p	P-value of the ~~biotc~~biotic slope.
r2	R^2 of the biological segment model.
adj_r2	Adjusted R^2 of the biotic segment model.
sse	Sum of squared errors for the biotic segment model.
breakpoint_year	Breakpoint year if segmentation was accepted.
segmentation_accepted	Indicates whether two-segment segmentation was accepted.
segmentation_reason	Reason why segmentation was accepted or rejected.
selection_score_single	Selection score of the single-segment model. Represents the BIC score. Lower values indicate a better model.
selection_score_two	Selection score of the two-segment model. Represents the BIC score. Lower values indicate a better model.
bic_single	Bayesian Information Criterion (BIC) score of the single-segment model. Lower values indicate a better balance between model fit and complexity.
bic_two	Bayesian Information Criterion (BIC) score of the two-segment model. Lower values indicate a better balance between model fit and complexity.

bic_two	Bayesian Information Criterion (BIC) score of the two-segment model. Lower values indicate a better balance between model fit and complexity.
bic_delta	Difference between the two-segment and single-segment BIC scores (bic_two − bic_single). Negative values favour segmentation, whereas positive values favour the single-segment model.
r2_gain	Improvement in adjusted R^2 from segmentation.
bio_y_start_hat	Predicted biotic value at the start of the segment.
bio_y_end_hat	Predicted biotic value at the end of the segment.
bio_duration_years	Duration of the biotic segment in years.
bio_abs_change	Absolute biotic change across the segment.
bio_rate_abs_per_year	Absolute biotic rate-of-change per year.
bio_boundary_max	Larger of predicted start and end biotic values, used for normalization.
bio_rate_pct_per_year	Normalized biotic rate-of-change (% yr^{-1}). Main biotic rate metric.
bio_total_pct_change	Total normalized biotic change across the segment (%).
abio_var	Abiotic variable paired with the biotic segment.
abio_title	Human-readable abiotic variable name.
abio_ylab	Y-axis label used for abiotic plots.
abio_n_months	Number of monthly abiotic observations used for trend estimation.
abio_analysis_start	First year of the abiotic analysis window.
abio_analysis_end	Last year of the abiotic analysis window.
abio_y_start_hat	Predicted abiotic value at the start of the abiotic analysis period.
abio_y_end_hat	Predicted abiotic value at the end of the abiotic analysis period.
abio_abs_change	Total abiotic change across the analysis period.
abio_rate_abs_per_year	Abiotic rate-of-change in native units per year. Main abiotic rate metric.
abio_slope_se	Standard error of the abio trend slope.
abio_slope_p	P-value of the abiotic trend slope.
abio_r2	R^2 of the abiotic trend model.
abio_adj_r2	Adjusted R^2 of the abiotic trend model.
abio_status	Status of abiotic trend estimation.
TDT	Trend Detection Time estimate.

uTDT	Uncertainty-adjusted Trend Detection Time estimate.
omega0	Abiotic trend magnitude used in TDT calculation.
sigmaN	Noise estimate used in TDT calculation.
phi	Lag-1 autocorrelation of abiotic trend residuals.
tdt_n	Number of observations used for TDT estimation.
tdt_qc	Quality-control flag for TDT estimate.
tdt_qc_reason	Explanation of the TDT quality-control flag.

VII. TESTING AND USER FEEDBACK

Independent testing with diverse biological datasets and users of different technical backgrounds is an essential part of validating and improving the workflow. Users are encouraged to test the tool with real datasets that include different biological organisation types, temporal resolutions, taxonomic structures, missing-data patterns, and documented method changes.

VIII. LITERATURE

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